Type MS · Using the sine rule, including the two-answer case

Answer 1 of 14 · Question 3 part whole · 5 marks

2024 NOV · Paper 01 · extract 1 of 12 · find part whole in this mark scheme extract

3	$\frac{\sin 25}{7}=\frac{\sin B}{10}$ oe $(B=)\sin^{-1}\left(\frac{10\sin 25}{7}\right)$ oe $(B=37.138(254541)$ or $142.861(7455))$	M1dM1
	$(C=180-[(180-37.138(25454))+25])$	
	$(C=)12.138(25451)$	A1 (M1 on ePen)
	$(c^2=)7^2+10^2-2(7)(10)\cos"12.138(25451)"$ or	
	$\frac{c}{\sin"12.138(25451)"}=\frac{7}{\sin 25}$ oe eg $(c=)\frac{7}{\sin 25}\times\sin"12.138(25451)"$ or	
	$\frac{c}{\sin"12.138(25451)"}=\frac{10}{\sin"142.861(7455)"}$ oe	
	eg $(c=)\frac{10}{\sin"142.861(7455)"}\times\sin"12.138(25451)"$	
	**allow students to follow through using their acute angle for B	ddM1
	$(c^2=)7^2+10^2-2(7)(10)\cos"117.861(7455)"$ or	
	$\frac{c}{\sin"117.861(7455)"}=\frac{7}{\sin 25}$ oe eg $(c=)\frac{7}{\sin 25}\times\sin"117.861(7455)"$ or	
	$\frac{c}{\sin"117.861(7455)"}=\frac{10}{\sin"37.138(254541)"}$ oe	
	eg $(c=)\frac{10}{\sin"37.138(254541)"}\times\sin"117.861(7455)"$	
	$(c=)3.5(\text{cm})$	A1 [5]
ALT1	$7^2=10^2+(AB)^2-2(10)(AB)\cos 25$	M1
	$0=(AB)^2-20\cos 25(AB)+100-49$ $(20\cos 25=18.12615574)$	dM1
	$0=(AB)^2-20\cos 25(AB)+51$	A1 (M1 on ePen)
	$(AB=)\frac{"--20\cos 25"\pm\sqrt{("-20\cos 25")^2-4("1")("51")}}{2\times"1"}(=3.48, 14.64)$	ddM1
	$(c=)3.5(\text{cm})$	A1 [5]
ALT2	M1 M1 as main scheme (then students are working in right angled triangles).	
	$(AX=)10\cos 25$ or $9.(06307787)$	B1 (M1 on ePen)
	$(BX=)7\sin("37.14.....")\Rightarrow 10\cos 25-7\sin("37.14.....")$	
	or if working with acute angle	ddM1
	$(BX=)7\sin("37.14.....")\Rightarrow 10\cos 25+7\sin("37.14.....")$	
	$=3.5((cm))$	A1 [5]
		Total 5 marks

Type MS · Using the sine rule, including the two-answer case

Answer 1 of 14 · Question 3 part whole · 5 marks

2024 NOV · Paper 01 · extract 2 of 12 · find part whole in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For correctly taking out 4 as a factor (must be explicitly seen **as in the mark scheme**).
	A1*cso	For the correct answer, minimum steps as shown, **no errors seen or omissions**.
(b)	M1	For an attempt to find the binomial expansion for the given expression. • The expansion must begin with 1 • The denominators must be correct (ie. 2! And 3! oe) on the third and fourth terms. • The power of x must be correct (ie. Must see $\frac{bx}{4}, \left(\frac{bx}{4}\right)^2$ and $\left(\frac{bx}{4}\right)^3$ with the correct corresponding denominators). **Simplification not necessary.**
	M1	For any two terms correct and unsimplified in the binomial expansion. It is not necessary to see the term in a.
	A1	For a fully correct unsimplified binomial expansion. It is not necessary to see the term in a.
	A1*cso	For the correct P, Q, R and S, stated explicitly or embedded, with no omissions or errors seen.
It is not necessary to see this step to gain the mark, but if seen, must be checked to ensure no errors: $\left(\frac{a}{2}\right)\left[1+\left(-\frac{bx}{8}\right)+\frac{3}{128}b^2x^2+\left(-\frac{5}{1024}b^3x^3\right)\right]$ oe As must any intermediate steps, as this is a show question.		
(c)	M1	For $(Q) = \frac{128}{5}(S)$ or $\frac{5}{128}(Q) = (S)$ in terms of a and b.
	A1*	For b correct, minimum steps as shown, no errors or omissions.
	B1*cso	For a correct equation as shown leading to the correct value of a, no errors or omissions. Although B marks are independent of method, this is a show question and this mark won't be awarded if there are errors in the work.
(d)	M1	For equating either form to $\frac{\sqrt{6}}{2}$ or $\frac{3}{\sqrt{6}}$, leading to a value for x (ie an incorrect rearrangement may be seen, but candidate must reach a value of x)
	dM1	For correctly substituting their value of x into their expansion with the given values of a and b, the given expressions for Q and S and their expressions for P and R Dependent on previous method mark. 1.207 will imply this mark.
	A1	For awrt 1.207 (note calculator value is 1.2247…)

Type MS · Using the sine rule, including the two-answer case

Answer 1 of 14 · Question 3 part whole · 5 marks

2024 NOV · Paper 01 · extract 3 of 12 · find part whole in this mark scheme extract

Question number	Scheme	Marks
7 (a)	$\frac{1}{4}\pi r^2 + 2xr = 100 \quad \left(100 - \frac{1}{4}\pi r^2 = 2xr\right) \Rightarrow x = \frac{50}{r} - \frac{1}{8}\pi r$ oe for example $\frac{100 - \frac{\pi r^2}{4}}{2r}$ $\frac{400 - \pi r^2}{8r}$	M1
	$(P =)\frac{1}{4}(2\pi r) + 4x + 2r$ oe for example $(P =)\frac{1}{2}\pi r + 2x + 2x + 2r$	M1
	$(P =)\frac{1}{2}\pi r + 4\left(\frac{50}{r} - \frac{1}{8}\pi r\right) + 2r \left(= \frac{1}{2}\pi r + \frac{200}{r} - \frac{1}{2}\pi r + 2r\right)$ oe	M1
	$P = \frac{200}{r} + 2r$ *	A1*cso [4]
(b)	$\left(\frac{dP}{dr} =\right) -\frac{200}{r^2} + 2$	M1
	$0 = "-\frac{200}{r^2} + 2" \Rightarrow r =$	M1
	$(r =)10$	A1
	$\left(\frac{d^2P}{dr^2} =\right)\frac{400}{r^3}$	M1
	$r = 10, \frac{d^2P}{dr^2} > 0$ therefore minimum or $\frac{d^2P}{dr^2} = \frac{400}{10^3}$ or $\frac{2}{5} > 0$ therefore minimum	A1 [5]
(c)	$P = \frac{200}{("10")} + 2("10") = 40$	M1 A1 [2]
		Total 11 marks

Type MS · Using the sine rule, including the two-answer case

Answer 1 of 14 · Question 3 part whole · 5 marks

2024 NOV · Paper 01 · extract 4 of 12 · find part whole in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	Writes the correct (unsimplified) equation for the area and rearranges to give $x =$ The rearrangement may contain up to 2 errors and does not have to be simplified.
	M1	Writes the correct formula or expression for the perimeter. It is not necessary to see $P =$ at this point and the expression does not need to be simplified.
	M1	Correctly substitutes their expression for x into their formula for the perimeter Not a dependent mark **but** ***x* must be of the form** $\frac{a}{r}+b\pi r \quad a,b\neq 0$ ***P* must be of the form** $d\pi r+ex+fr \quad d,e,f\neq 0$ Not necessary to see $P =$ for this mark.
	A1*cso	For the given result, minimum steps as shown, with no errors seen. It is not necessary to see the steps in brackets. They are present to help examiners check intermediate work as there shouldn't be incorrect work for this mark. It is necessary to see $P =$ at some point for this mark to be awarded.
(b)	M1	For attempting to differentiate P wrt r At least one term must be fully correct, the other – follow general guidance.
	M1	For setting their derivative (must involve a changed expression and have 2 terms) = 0 and a completely correct rearrangement to find the value of r
	A1	For ($r =$) 10, ignore $r = -10$
	M1	For an attempt to find the second derivative to give an expression of the form $\pm\frac{a}{r^3}$
	A1	For a correct second derivative and correct justification that this is a minimum, with some form of conclusion. All work to be correct for this mark. i.e it is acceptable, as this is a simple substitution to state (if $r = 10$) $\frac{\mathrm{d}^2P}{\mathrm{d}r^2}>0$, but if a substitution of $r = 10$ is made, the value for the second derivative must be given as 0.4 Work with $r = -10$ to find a maximum may be ignored.
ALT final A1	A1	Testing and substituting into the correct first derivative with appropriate values either side of $r = 10$. For correct justification that this is a minimum, with some form of conclusion. All work to be correct for this mark.
(c)	M1	For correct substitution of their r into the given formula for P.
	A1	40

Type MS · Using the sine rule, including the two-answer case

Answer 1 of 14 · Question 3 part whole · 5 marks

2024 NOV · Paper 01 · extract 5 of 12 · find part whole in this mark scheme extract

Question number	Scheme	Marks
8 (i) (a)	$\left(\tan 2A = \tan(A+A) =\right)\dfrac{\tan A+\tan A}{1-\tan A\tan A}$ $=\dfrac{2\tan A}{1-\tan^2 A}$ *	M1 A1cso* [2]
(b)	$\tan A-\dfrac{2\tan A}{1-\tan^2 A}=0$ $\tan A(1-\tan^2 A)-2\tan A=0$ $\tan A-\tan^3 A-2\tan A=0$ $\tan^3 A+\tan A=0$ $\tan A(\tan^2 A+1)=0$ $\tan A=0 \qquad A=0, 180$ $\left(\tan^2 A=-1 \qquad \text{no solutions}\right)$	M1 M1 M1 A1, A1 [5]
ALT	$\tan A-\dfrac{2\tan A}{1-\tan^2 A}=0$ $\tan A\left(1-\dfrac{2}{1-\tan^2 A}\right)=0 \Rightarrow 1-\tan^2 A-2=0 \text{ or } 2=1-\tan^2 A$ $\tan^2 A=-1$ $\tan A=0 \qquad A=0, 180$ $\left(\tan^2 A=-1 \qquad \text{no solutions}\right)$	
(ii)	$\cos x\cos\dfrac{\pi}{6}+\sin x\sin\dfrac{\pi}{6}\left(=\sin x\right)$ $\sin x+\sqrt{3}\cos x=2\sin x \Rightarrow \sqrt{3}=\tan x$ $x=-\dfrac{2\pi}{3},\dfrac{\pi}{3},\dfrac{4\pi}{3}$	M1 dM1 A1 A1 [4]
		Total 11 marks

Type MS · Using the sine rule, including the two-answer case

Answer 1 of 14 · Question 3 part whole · 5 marks

2024 NOV · Paper 01 · extract 6 of 12 · find part whole in this mark scheme extract

Part	Mark	Additional Guidance
(i) (a)	M1	For $\dfrac{\tan A+\tan A}{1-\tan A\tan A}$
	A1cso*	For the given result shown with minimum steps as shown in the MS, no errors or omissions seen. Extra steps must be correct and checked.
(b)	M1	For the correct substitution of the given result from part (a).
	M1	For correctly multiplying by $1-\tan^2 A$ or for sight of a correct single fraction (which could possibly afterwards be simplified incorrectly) and elimination of $1-\tan^2 A$ as the denominator.
	M1	For a correct rearrangement of their equation, factorising correctly, leading to a quadratic factor and a linear factor = 0
	A1	For $A = 0$ or 180 must follow M3
	A1	For $A = 0$ & 180 with no other values in range. Ignore values outside range. Must follow M3 Note if you see any solutions based on trial and improvement, for this question, this will score 0 marks.
	eg $\dfrac{\tan A-\tan^3 A-2\tan A}{1-\tan^2 A}=0$ is M1 M1 M0 $3\tan A-\tan^3 A=0$	
ALT	M1	For the correct substitution of the given result from part (a).
	M1	For a correct factorisation and correctly multiplying by $1-\tan^2 A$
	M1	For a correct rearrangement of their equation leading to $\tan^2 A = P \qquad P \neq 0$
	A1	For $A = 0$ or 180 must follow M3
	A1	For $A = 0$ & 180 with no other values in range. Ignore values outside range. Must follow M3 Note if you see any solutions based on trial and improvement, for this question, this will score 0 marks.
(ii)	M1	For correctly using the formula for cos $(A - B)$ as shown.
	dM1	For simplifying and rearranging to tan $x = a$, allow errors in rearrangement, a must be an exact value. Dependent on previous method mark.
	A1 A1	A1 for one value or for any correct value(s) with extra value in range. A2 for all three values with no extra values in range. Ignore extra values out of range correct or otherwise.

Type MS · Using the sine rule, including the two-answer case

Answer 1 of 14 · Question 3 part whole · 5 marks

2024 NOV · Paper 01 · extract 7 of 12 · find part whole in this mark scheme extract

Question number	Scheme	Marks
9	$\left(x=y^2+1 \qquad x=4-2y\right) \quad \left(y=\sqrt{x-1} \quad y=\dfrac{4-x}{2}\right)$	
	$4-2y=y^2+1 \qquad \text{or} \qquad \sqrt{x-1}=\dfrac{4-x}{2}$	M1
	$0=y^2+2y-3 \qquad 0=x^2-12x+20$	A1
	$0=(y+3)(y-1) \qquad 0=(x-2)(x-10)$	M1
	$\left(y=1,\right)x=2 \qquad x=2$	A1
	$\left(y^2=\left(2-\dfrac{x}{2}\right)^2=4-2x+\dfrac{x^2}{4}\right)$	
	$\left(0=x-1\right)\Rightarrow x=1 \qquad \left(2(0)+x-4=0\right)\Rightarrow x=4$	B1B1
	$\pi\int_{"-1"}^{"2"}(x-1)\mathrm{d}x+\pi\int_{"-2"}^{"4"}\left(2-\dfrac{x}{2}\right)^2\mathrm{d}x \text{ or } \pi\int_{"-1"}^{"2"}(x-1)\mathrm{d}x+\dfrac{1}{3}\pi\times"1"^2\times("4"-"2")^2$	ddM1
	$\left[\pi\int_1^2(x-1)\,\mathrm{d}x+\pi\int_2^4\left(4-2x+\dfrac{x^2}{4}\right)\mathrm{d}x \text{ or } \dfrac{1}{3}\pi\times1^2\times2\right]$	
	$(\pi)\left[\dfrac{x^2}{2}-x\right]_{(1)}^{(2)}+(\pi)\left[\left(4x-\dfrac{2x^2}{2}+\dfrac{x^3}{12}\right)\right]_{(2)}^{(4)}$	M1
	$(\pi)\left[\left(\dfrac{"2"^2}{2}-"2"\right)-\left(\dfrac{"1"^2}{2}-"1"\right)\right]+(\pi)\left[\left(4("4")-\dfrac{2("4")^2}{2}+\dfrac{"4"^3}{12}\right)-\left(4("2")-\dfrac{2("2")^2}{2}+\dfrac{"2"^3}{12}\right)\right]$	M1
	$\left(\dfrac{1}{2}\pi+\dfrac{2}{3}\pi\right)$	
	$=\dfrac{7}{6}\pi$ oe	A1 [10]
		Total 10 marks

Type MS · Using the sine rule, including the two-answer case

Answer 1 of 14 · Question 3 part whole · 5 marks

2024 NOV · Paper 01 · extract 8 of 12 · find part whole in this mark scheme extract

Part	Mark	Additional Guidance
	M1	For correctly placing the 2 equations equal to each other and any attempt to form a quadratic = 0 in x or y This attempt doesn't need to be correct, but they must reach an unsimplified quadratic
	A1	For the correct 3TQ = 0
	M1	For any method to solve their quadratic – see general guidance for minimally acceptable attempt. $x = 2$ will imply this mark.
	A1	For $x = 2$
	B1	For $x = 1$ or $x = 4$
	B1	For $x = 1$ and $x = 4$
	ddM1	For $\pi\int_{"1"}^{"2"}(x-1)\mathrm{d}x + \pi\int_{"2"}^{"4"}\left(2-\frac{x}{2}\right)^2 \mathrm{d}x$ or $\pi\int_{"1"}^{"2"}(x-1)\mathrm{d}x + \frac{1}{3}\pi\times"1"^2\times("4"-"2")^2$ Dependent on both previous method marks. Also dependent on having attempted to find the intersection of the curve and the line with the x-axis. Allow their "1" and their "4" limits as long as it's clear the candidate has attempted to find the intersections of the **line and curve** with the x axis
	M1	For integrating (minimum attempt – see general guidance and also, no power of x to decrease) their terms in their volume of revolution. There must be a minimum of 2 terms. This is not a dependent mark, the mark is given for integration, π and limits **do not need to be present** for this mark to be awarded. If students have written an incorrect formula for the volume of the cone, this mark may still be awarded for integrating. If expressions have been combined incorrectly, this mark for integration of a minimum of 2 terms may be given.
	M1	For correct substitution of limits into any changed expression. Also not dependent, any substitution of what candidates consider to be their limits into their changed expression for the volume of revolution may be awarded this mark. If students have written an incorrect formula for the volume of the cone, this mark may still be awarded for integrating. If expressions have been combined incorrectly, this mark is for substitution of limits into any changed expression(s), minimum 2 terms, with each limit substituted correctly at least once and may be given.
	A1	For $\frac{7}{6}\pi$ oe

Type MS · Using the sine rule, including the two-answer case

Answer 1 of 14 · Question 3 part whole · 5 marks

2024 NOV · Paper 01 · extract 9 of 12 · find part whole in this mark scheme extract

Question number	Scheme	Marks
10 (a)	$\left(3-9\left(\frac{4}{9}x+x^2\right)\right.$ or $\left.-9\left(-\frac{1}{3}+\frac{4}{9}x+x^2\right)\right)$ $3-9\left[\left(x+\frac{2}{9}\right)^2-\frac{4}{81}\right]$ or $-9\left[\left(x+\frac{2}{9}\right)^2-\frac{4}{81}-\frac{1}{3}\right]$ or $-1\left[9\left(x+\frac{2}{9}\right)^2-\frac{36}{81}-3\right]$ or $3+(1)\left[-9\left(x+\frac{2}{9}\right)^2+\frac{36}{81}\right]$ $\frac{31}{9}-9\left(x+\frac{2}{9}\right)^2$ $A=\frac{31}{9}$ $B=9$ $C=\frac{2}{9}$ oe	M1 A1 A1 A1 [4]
(b)	$"\frac{31}{9}"$	B1ft [1]
(c)	$\alpha+\beta=-\frac{4}{9}$ $\alpha\beta=-\frac{3}{9}$ oe $\frac{3\alpha}{\beta}+\frac{3\beta}{\alpha}\left(=\frac{3\left(\alpha^2+\beta^2\right)}{\alpha\beta}\right)=\frac{3\left((\alpha+\beta)^2-2\alpha\beta\right)}{\alpha\beta}$ $\frac{3\left[\left("-\frac{4}{9}"\right)^2-2\left("-\frac{3}{9}"\right)\right]}{"-\frac{3}{9}"}$ $\left(=-\frac{70}{9}\right)$ $\left(\frac{3\alpha}{\beta}\times\frac{3\beta}{\alpha}=\right)9$ $x^2+"\frac{70}{9}"x+"9"(=0)$ $9x^2+70x+81=0$	B1 M1 A1ft B1 M1 A1cso [6]
(d)	$\left((x+y)^3=\right)x^3+3x^2y+3xy^2+y^3=x^3+y^3+3xy(x+y)$	B1*cso [1]
(e)	$(\alpha^2-\beta+\beta^2-\alpha=)$ $(\alpha+\beta)^2-2\alpha\beta-(\alpha+\beta)$ $\left((\alpha+\beta)^2=(\alpha+\beta)^2-2\alpha\beta=\frac{70}{81}\right)$ $\left("-\frac{4}{9}"\right)^2-2\left("-\frac{3}{9}"\right)-\left("-\frac{4}{9}"\right)$ $\left(=\frac{106}{81}\right)$ **candidates may also transfer a value for $\alpha^2+\beta^2$ from working in part (c)** $\left[(\alpha^2-\beta)(\beta^2-\alpha)=\alpha^2\beta^2-\beta^3-\alpha^3+\alpha\beta\right]$ $=(\alpha\beta)^2+\alpha\beta-(\alpha^3+\beta^3)$ or $(\alpha\beta)^2+\alpha\beta-\alpha^3-\beta^3$ $\left[(\alpha+\beta)^3=\alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3=\alpha^3+\beta^3+3\alpha\beta(\alpha+\beta)\right]$ $=(\alpha\beta)^2+\alpha\beta-(\alpha+\beta)^3+3\alpha\beta(\alpha+\beta)$ or $(\alpha\beta)^2+\alpha\beta-\left((\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)\right)$ $\left[=\left("-\frac{3}{9}"\right)^2+\left("-\frac{3}{9}"\right)-\left("-\frac{4}{9}"\right)^3+3\left("-\frac{3}{9}"\right)\left("-\frac{4}{9}"\right)\left(=\frac{226}{729}\right)\right]$ $"\frac{106}{81}"=-\frac{q}{3}$ or $"\frac{226}{729}"=\frac{r}{3}$ $q=-\frac{106}{27}$ and $r=\frac{226}{243}$	M1 A1ft M1 M1 dddM1 A1 [6]
		Total 18 marks

Type MS · Using the sine rule, including the two-answer case

Answer 1 of 14 · Question 3 part whole · 5 marks

2024 NOV · Paper 01 · extract 10 of 12 · find part whole in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For a method to complete the square to achieve as a minimum $3\pm9\left[\left(x\pm\frac{2}{9}\right)^2\pm p\right]$ or $\pm9\left[\left(x\pm\frac{2}{9}\right)^2\pm q\right]$ or $\pm1\left[9\left(x\pm\frac{2}{9}\right)^2\pm r\right]$ or $3\pm(1)\left[-9\left(x\pm\frac{2}{9}\right)^2+s\right]$ where p, q r or s are constants > 0.
	A1	For one of A, B or C correct.
	A1	For two of A, B or C correct.
	A1	For all of A, B or C correct For all accuracy marks, A, B and C may be either explicitly stated or embedded and if embedded correctly then stated incorrectly, we may isw. Correct A, B and C will imply 4 marks. 1 or 2 values correct, will need M1 to be gained first.
ALT	M1	For an attempt to expand $A-B(x+C)^2$ **AND** equate coefficients to the given f $(x)\Rightarrow A-Bx^2-2BCx-BC^2=3-4x-9x^2$ Allow $A\pm Bx^2\pm2BCx\pm BC^2$ for the expansion of $A-B(x+C)^2$ Must be an attempt to correctly equate at least one coefficient using **their** expansion. Eg $-B=-9\Rightarrow B=...$ $-2BC=-4\Rightarrow C=...$ $A-BC^2=3\Rightarrow A=...$
	A1A1A1	As main scheme.
(b)	B1ft	For **their** $\frac{31}{9}$ Note, strict 'hence' question, if $\frac{31}{9}$ is written with incorrect work in (a), B0. It is the candidate's final answer in part (a) which will be used to mark part (b) ft must be from $A-B(x+C)^2$ If $\frac{31}{9}$ is written with no work in (a), award the mark.
(c)	B1	Correct values for $\alpha+\beta$ **and** $\alpha\beta$. Award if and look for values not seen explicitly but embedded in the sum/product calculations for the new equation.
	M1	Reaches a **correct** expression ready for substitution of their values of $\alpha+\beta$ **and** $\alpha\beta$
	A1ft	Correctly substitutes their values for $\alpha+\beta$ **and** $\alpha\beta$ into a correct expression for the sum of roots. Simplification is not necessary.
	B1	For 9
	M1	Use $x^2-(\text{their sum of their roots})x+\text{their product of roots}$ "= 0" may be missing. This mark is not dependent and may be awarded for a clear substitution of anything we can see is their product and sum of roots.
	A1cso	For the given equation oe with integer coefficients. Must have = 0.
Note. It is possible to get the correct equation from using $\alpha+\beta=+\frac{4}{9}$ This will always achieve final A0 as it's a correct equation from incorrect working.		

Type MS · Using the sine rule, including the two-answer case

Answer 1 of 14 · Question 3 part whole · 5 marks

2024 NOV · Paper 01 · extract 11 of 12 · find part whole in this mark scheme extract

(d)	B1*cso	Complete and full algebra to show the given identity. Minimum steps as shown in MS, no errors or omissions. Extra steps checked. For this question, where students have chosen to do full expansion of brackets, we will allow them to recover brackets if following work is correct.
(e)	Look carefully for students transferring their work in (c) on $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$ as this can get the first 2 marks in (e), implied M1, A1ft. This work must be correct for their original sum and product of roots and must be shown. Examiners are not expected to check work if their substitution isn't shown.	
	M1	Reaches a **correct** expression ready for substitution of their values of $\alpha+\beta$ **and** $\alpha\beta$
	A1ft	Correctly substitutes their values for $\alpha+\beta$ **and** $\alpha\beta$ into a correct expression for sum of roots.
	M1	For the correct expression shown. Any equivalent, ready for substitution of $\alpha\beta$
	M1	For the correct expression shown. Any equiv ready for sub of $\alpha\beta$ and $\alpha+\beta$ For this mark and the previous M mark, we're looking for expressions preparing for substitution of candidates' values. You may see completion of this work in stages, eg when the values are subbed in later. Eg, you may see $\alpha^2\beta^2+\alpha\beta-(\alpha^3+\beta^3)\rightarrow\alpha^2\beta^2+\alpha\beta-(\alpha+\beta)^3+3\alpha\beta(\alpha+\beta)$ then $\left("-\frac{3}{9}"\right)^2+\left("-\frac{3}{9}"\right)-\left("-\frac{4}{9}"\right)^3+3\left("-\frac{3}{9}"\right)\left("-\frac{4}{9}"\right)$ which is M2, $(\alpha\beta)^2$ is implied.
	dddM1	For equating their sum of roots to $-\frac{q}{3}$ or their product of roots to $\frac{r}{3}$. Dep on all method marks. This mark may sometimes be implied by a correct p or q
	A1	For both correct values for q and r

In general, A marks (or M marks) cannot come from incorrect working (though a correct method followed by incorrect simplification can usually go on to gain M marks).

In this question, you are looking for any of the forms given in the notes, at any point (and disregarding incorrect work around it) to award M1.

Only when this is awarded, you may award following A marks. The exception being a fully correct answer may be awarded 4 marks without M1 being present.

Type MS · Using the sine rule, including the two-answer case

Answer 1 of 14 · Question 3 part whole · 5 marks

2024 NOV · Paper 01 · extract 12 of 12 · find part whole in this mark scheme extract

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Type MS · Using the sine rule, including the two-answer case

Answer 2 of 14 · Question 3 part b.i · 3 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 3 · find part b.i in this mark scheme extract

Question	Scheme	Marks
3(a)	$BC^2 = 10^2 + 8^2 - 2 \times 10 \times 8 \times \cos 100^\circ$ $BC = 13.848\ldots \approx 13.8$	M1 A1 [2]
(b)	(i) $\sin ABC = \frac{8\sin 100^\circ}{13.848\ldots} \Rightarrow ABC = 34.6752\ldots^\circ \approx 34.7^\circ$ (ii) $\angle ACB = 180^\circ - 100^\circ - 34.6752\ldots^\circ = 45.324\ldots^\circ \approx 45.3^\circ$	M1A1 B1FT [3]
	(i)(ii) $\cos ACB = \frac{8^2+13.848\ldots^2-10^2}{2\times 8\times 13.848\ldots} \Rightarrow ACB = 45.330\ldots$ $ABC = 180° - 100° - 45.330\ldots = 34.6699\ldots$	[M1A1 B1FT]
(c)	$\angle MBC = 34.675 \div 2 = 17.337\ldots^\circ$ and $\angle BMC = 180^\circ - 17.337^\circ = 117.33\ldots^\circ$ $MC = \frac{13.848\sin 17.337^\circ}{\sin 117.33^\circ} = 4.6293\ldots$ Area of BMC = $\frac{1}{2} \times 4.629 \times 13.848 \times \sin 45.324^\circ = 22.79\ldots \approx 22.8\ (\text{cm}^2)$	B1FT M1 M1A1 [4]
	ALT	
	$ABM = 34.675 \div 2 = 17.337\ldots^\circ$ and $AMB = 180 - 100 - 17.33\ldots = 62.66\ldots$ $BM = \frac{10\sin 100}{\sin 62.66\ldots} = 11.086\ldots$ Area of BMC = $\frac{1}{2} \times 13.848\ldots \times 11.086\ldots \times \sin 17.337\ldots = 22.87\ldots$ $\approx 22.9\ (\text{cm}^2)$	[B1FT M1 M1A1]
		Total 9 marks

Type MS · Using the sine rule, including the two-answer case

Answer 2 of 14 · Question 3 part b.i · 3 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 3 · find part b.i in this mark scheme extract

ALT		
(b)(i)(ii)	M1	For applying cosine rule correctly to find angle ACB Accept any appropriate trigonometry.
	A1	For awrt 45.3° (angle ACB)
	B1FT	FT $180 - 100 - their$ 45.330 ... (angle ABC)
(c)	B1ft	For both angles MBC and BMC
	M1	For applying the sine rule using their angles and BC to find length MC Allow alternative correct methods to find MC.
	M1	For using Area = $\frac{1}{2}ab\sin C$ using their length and angles. $Their\ MC \neq 4$
	A1	Awrt 22.8 or awrt 22.9
ALT		
(c)	B1ft	For both angles ABM and AMB
	M1	For applying sine rule using their angles to find length BM. Allow alternative correct methods to find BM.
	M1	For using Area = $\frac{1}{2}ab\sin C$ using their length and angles. $Their\ MC \neq 4$ if used.
	A1	Awrt 22.8 or awrt 22.9

Type MS · Using the sine rule, including the two-answer case

Answer 2 of 14 · Question 3 part b.i · 3 marks

2023 JUN · Paper 01 (regional R) · extract 3 of 3 · find part b.i in this mark scheme extract

Question	Scheme	Marks

Type MS · Using the sine rule, including the two-answer case

Answer 3 of 14 · Question 3 part c.i · 0 marks

2024 JUN · Paper 01 (regional R) · extract 1 of 5 · find part c.i in this mark scheme extract

		$\left(-\frac{}{3n}\right) \Rightarrow \left(\frac{}{9n}\right)$ or $\left(-\frac{}{3n}\right) \Rightarrow -\left(\frac{}{9n^2}\right)$ then this is M0 Their equation must be either a quadratic or linear if they cancel an n This is dependent on the previous M mark only.
	A1	For the correct value of n **or** A $\quad n=-\frac{2}{3}, \quad A=\frac{1}{2}$ isw $n = 0$ seen. This is equivalent to cancelling n's
	A1	For the correct value of n **and** A
	ALT 1	**First two marks same as main method**
	M1	Substitutes n in terms of A into an equation in A $\frac{'-\frac{1}{3A}'\left('-\frac{1}{3A}'-1\right)(A)^2}{2}=\frac{5}{36}$ **Do not** allow a substitution of $n=-\frac{1}{3}$ for this mark
	dM1	For solving **their** equation in A to find a value for A Their equation must be either a quadratic or linear if they cancel an A This is dependent on the previous M mark only.
	A1	For the correct value of n **or** A $\quad n=-\frac{2}{3}, \quad A=\frac{1}{2}$
	A1	For the correct value of n **and** A
	ALT 2	**First two marks same as main method**
	M1	Substitutes nA into $\frac{(nA)^2-A(An)}{2!}=\frac{5}{36}$
	dM1	Solves their equation to find a value for A provided no errors introduced. This is dependent on the previous M mark only.
	A1	For the correct value of n **or** A $\quad n=-\frac{2}{3}, \quad A=\frac{1}{2}$
	A1	For the correct value of n **and** A

Type MS · Using the sine rule, including the two-answer case

Answer 3 of 14 · Question 3 part c.i · 0 marks

2024 JUN · Paper 01 (regional R) · extract 2 of 5 · find part c.i in this mark scheme extract

(b)	M1	Uses the **correct form** for the fourth term of a binomial expansion with their A and their n. You may see this in terms of A and n in part (a) Accept as a minimum, the correct power of $\left(\frac{1}{2}\right)^3$ with the correct denominator i.e. 3! Allow the presence of x^3 for this mark.
	A1	For the correct value of $-\frac{5}{81}$ If you see the correct value following the correct values of A and n with no working, award M1A1

Type MS · Using the sine rule, including the two-answer case

Answer 3 of 14 · Question 3 part c.i · 0 marks

2024 JUN · Paper 01 (regional R) · extract 3 of 5 · find part c.i in this mark scheme extract

Question	Scheme	Marks
3(a)	$36=\frac{1}{2}\times x\times(x+7)\times\sin 30^{\circ}\Rightarrow 36=\frac{x(x+7)}{4}\Rightarrow x^2+7x-144=0$	M1
	[Accept also e.g., $x^2+7x=144$]	
	$(x-9)(x+16)=0\Rightarrow x=9,(-16)\,(\text{cm})$ *	M1A1 cso [3]
(b)	$BC=\sqrt{9^2+16^2-2\times9\times16\cos 30^{\circ}}=9.3586...\approx 9.36\,(\text{cm})$	M1A1 [2]
(c)(i)	$\frac{\sin ABC}{16}=\frac{\sin 30^{\circ}}{9.3586}\Rightarrow \angle ABC=180^{\circ}-58.7408...\approx 121.3^{\circ}$	M1A1
(ii)	$\angle BCA=180^{\circ}-30^{\circ}-121.259^{\circ}=28.740...\approx 28.7^{\circ}$	B1ft [3]
		Total 8 marks

Type MS · Using the sine rule, including the two-answer case

Answer 3 of 14 · Question 3 part c.i · 0 marks

2024 JUN · Paper 01 (regional R) · extract 4 of 5 · find part c.i in this mark scheme extract

Part	Mark	Notes
(a)	B1	States or uses $\theta = \frac{21-r}{r}$
	M1	Uses the correct formula for the area of a sector and substitutes their expression for θ
	A1 cso	For the correct expression for A exactly as written with no errors. That is $A = \frac{r}{2}(21-r)$ including $A =$.. However, if they have a string of working, then allow $A =$ at the top For example ., $A =$ ..... $=$.... $= \ldots\ldots\ldots\ldots = \frac{r}{2}(21-r)$ is acceptable
	ALT	
	B1	States $l = 21-r$
	M1	Uses the alternative formula $A = \frac{1}{2}rl$ and substitutes in their l
	A1 cso	For the correct expression for A exactly as written with no errors.
(b)	B1	Forms a correct inequality **and** a correct 3TQ (accept in any form) using the given expression for A and the value of 27
	M1	Solves their 3TQ by any method to obtain two values for r If they use a calculator to solve an incorrect 3TQ, award only if a method is seen. If the 3TQ is correct, and they use a calculator, award for correct roots seen.
	M1	Forms an inequality for the **inside** region using their values of r Allow any critical values which were obtained from their quadratic equation. Accept any acceptable form that defines a continuous **inside** region For example: $3 \leqslant r$ and $r \leqslant 18$ or $3 \leqslant r \cap r \leqslant 18$ Accept even $3 \leqslant r$ or $r \leqslant 18$
	A1	For the correct inequality only in this form. $3 \leqslant r \leqslant 18$
(c)	B1ft	For both $\theta = \frac{1}{6}$ and $\theta = 6$ You must follow through their values for r for this mark.
	B1	For the correct inequality only in this form.. $\frac{1}{6} \leqslant \theta \leqslant 6$

Type MS · Using the sine rule, including the two-answer case

Answer 3 of 14 · Question 3 part c.i · 0 marks

2024 JUN · Paper 01 (regional R) · extract 5 of 5 · find part c.i in this mark scheme extract

Question	Scheme	Marks
5(a)	**Mark both parts together**	

		$\left(-\frac{}{3n}\right) \Rightarrow \left(\frac{}{9n}\right)$ or $\left(-\frac{}{3n}\right) \Rightarrow -\left(\frac{}{9n^2}\right)$ then this is M0 Their equation must be either a quadratic or linear if they cancel an n This is dependent on the previous M mark only.
	A1	For the correct value of n **or** A $\quad n=-\frac{2}{3}, \quad A=\frac{1}{2}$ isw $n = 0$ seen. This is equivalent to cancelling n's
	A1	For the correct value of n **and** A
	ALT 1	**First two marks same as main method**
	M1	Substitutes n in terms of A into an equation in A $\frac{'-\frac{1}{3A}'\left('-\frac{1}{3A}'-1\right)(A)^2}{2}=\frac{5}{36}$ **Do not** allow a substitution of $n=-\frac{1}{3}$ for this mark
	dM1	For solving **their** equation in A to find a value for A Their equation must be either a quadratic or linear if they cancel an A This is dependent on the previous M mark only.
	A1	For the correct value of n **or** A $\quad n=-\frac{2}{3}, \quad A=\frac{1}{2}$
	A1	For the correct value of n **and** A
	ALT 2	**First two marks same as main method**
	M1	Substitutes nA into $\frac{(nA)^2-A(An)}{2!}=\frac{5}{36}$
	dM1	Solves their equation to find a value for A provided no errors introduced. This is dependent on the previous M mark only.
	A1	For the correct value of n **or** A $\quad n=-\frac{2}{3}, \quad A=\frac{1}{2}$
	A1	For the correct value of n **and** A

Type MS · Using the sine rule, including the two-answer case

Answer 4 of 14 · Question 3 part c.ii · 3 marks

2024 JUN · Paper 01 (regional R) · extract 2 of 5 · find part c.ii in this mark scheme extract

(b)	M1	Uses the **correct form** for the fourth term of a binomial expansion with their A and their n. You may see this in terms of A and n in part (a) Accept as a minimum, the correct power of $\left(\frac{1}{2}\right)^3$ with the correct denominator i.e. 3! Allow the presence of x^3 for this mark.
	A1	For the correct value of $-\frac{5}{81}$ If you see the correct value following the correct values of A and n with no working, award M1A1

Type MS · Using the sine rule, including the two-answer case

Answer 4 of 14 · Question 3 part c.ii · 3 marks

2024 JUN · Paper 01 (regional R) · extract 3 of 5 · find part c.ii in this mark scheme extract

Question	Scheme	Marks
3(a)	$36=\frac{1}{2}\times x\times(x+7)\times\sin 30^\circ \Rightarrow 36=\frac{x(x+7)}{4} \Rightarrow x^2+7x-144=0$	M1
	[Accept also e.g., $x^2+7x=144$]	
	$(x-9)(x+16)=0 \Rightarrow x=9,(-16)\,(\text{cm})$*	M1A1 cso [3]
(b)	$BC=\sqrt{9^2+16^2-2\times9\times16\cos30^\circ}=9.3586...\approx9.36\,(\text{cm})$	M1A1 [2]
(c)(i)	$\frac{\sin ABC}{16}=\frac{\sin30^\circ}{9.3586} \Rightarrow \angle ABC=180^\circ-58.7408...\approx121.3^\circ$	M1A1
(ii)	$\angle BCA=180^\circ-30^\circ-121.259^\circ=28.740...\approx28.7^\circ$	B1ft [3]
		Total 8 marks

Type MS · Using the sine rule, including the two-answer case

Answer 4 of 14 · Question 3 part c.ii · 3 marks

2024 JUN · Paper 01 (regional R) · extract 4 of 5 · find part c.ii in this mark scheme extract

Part	Mark	Notes
(a)	B1	States or uses $\theta = \frac{21-r}{r}$
	M1	Uses the correct formula for the area of a sector and substitutes their expression for θ
	A1 cso	For the correct expression for A exactly as written with no errors. That is $A = \frac{r}{2}(21-r)$ including $A =$.. However, if they have a string of working, then allow $A =$ at the top For example ., $A =$ ..... $=$.... $= \ldots\ldots\ldots\ldots = \frac{r}{2}(21-r)$ is acceptable
	ALT	
	B1	States $l = 21-r$
	M1	Uses the alternative formula $A = \frac{1}{2}rl$ and substitutes in their l
	A1 cso	For the correct expression for A exactly as written with no errors.
(b)	B1	Forms a correct inequality **and** a correct 3TQ (accept in any form) using the given expression for A and the value of 27
	M1	Solves their 3TQ by any method to obtain two values for r If they use a calculator to solve an incorrect 3TQ, award only if a method is seen. If the 3TQ is correct, and they use a calculator, award for correct roots seen.
	M1	Forms an inequality for the **inside** region using their values of r Allow any critical values which were obtained from their quadratic equation. Accept any acceptable form that defines a continuous **inside** region For example: $3 \leqslant r$ and $r \leqslant 18$ or $3 \leqslant r \cap r \leqslant 18$ Accept even $3 \leqslant r$ or $r \leqslant 18$
	A1	For the correct inequality only in this form. $3 \leqslant r \leqslant 18$
(c)	B1ft	For both $\theta = \frac{1}{6}$ and $\theta = 6$ You must follow through their values for r for this mark.
	B1	For the correct inequality only in this form.. $\frac{1}{6} \leqslant \theta \leqslant 6$

Type MS · Using the sine rule, including the two-answer case

Answer 4 of 14 · Question 3 part c.ii · 3 marks

2024 JUN · Paper 01 (regional R) · extract 5 of 5 · find part c.ii in this mark scheme extract

Question	Scheme	Marks
5(a)	**Mark both parts together**	

Type MS · Using the sine rule, including the two-answer case

Answer 5 of 14 · Question 2 part whole · 3 marks

2023 JAN · Paper 01 (regional R) · extract 1 of 2 · find part whole in this mark scheme extract

2	$\frac{\sin\angle BCA}{10}=\frac{\sin 50}{9}\Rightarrow\angle BCA=58.3381...^{\circ}\Rightarrow 58.3^{\circ},\ 121.7^{\circ}$	M1A1A1
		Total 3 marks

Question	Notes	Marks
2	Uses sine rule or any other appropriate trigonometry in triangle ABC $\frac{\sin\angle BCA}{10}=\frac{\sin 50}{9}$ Note: the perpendicular height of the triangle from B to AC is 7.66044 cm. Their method must be complete for the award of this mark.	M1
	$\angle BCA=58.3381...^{\circ}$	A1
	One possible value is awrt 58.3° and the other possible value is awrt 121.7°	A1 [3]
		Total 3 marks

Type MS · Using the sine rule, including the two-answer case

Answer 5 of 14 · Question 2 part whole · 3 marks

2023 JAN · Paper 01 (regional R) · extract 2 of 2 · find part whole in this mark scheme extract

Question	Scheme	Marks

Type MS · Using the sine rule, including the two-answer case

Answer 6 of 14 · Question 4 part a · 4 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

		For $t = \frac{}{4}$
(b)	M1	For setting $v = 0$ and attempting to solve, see general guidance for definition of a valid attempt to solve.
	A1	For one correct value of either T_1 or T_2
	A1	For both correct values
(c)	**Note:** **The question states 'Show your working clearly' No integration seen – award M0A0M0A0**	
	M1	For an attempt to integrate, see general guidance.
	A1	For a fully correct integration (constant of integration need not be shown).
	M1	For correct substitution of **their** T_1 and T_2 **into a changed expression AND subtracts their values correctly.** If these are not the correct values, then must see each limit correctly substituted into their expression at least once, If the correct values are used, but their final answer is not correct, explicit substitution must be seen for the award of this mark. **Note:** Mark may be implied by a correct answer given **following correct integration**
	A1	Correct answer **which must be a positive value.**

Type MS · Using the sine rule, including the two-answer case

Answer 6 of 14 · Question 4 part a · 4 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks
4 (a)	$\frac{\sin "A"}{5.9}=\frac{\sin 29}{4.3}$ oe	M1
	$(A=)\sin^{-1}\left(\frac{\sin 29}{4.3}\times 5.9\right)$ or $\sin^{-1}("0.66520.....")$ oe	dM1
	$A=41.7$	A1
	$A=180-"41.69796....."=138.3$	A1
		[4]
(b)	$180-(29+"138.30203")$ $(=12.69796.....)$	M1
	$(AC)^2=5.9^2+4.3^2-2\times 5.9\times 4.3\cos(180-(29+"138.30203"))$ or	M1
	$AC=\sqrt{5.9^2+4.3^2-2\times 5.9\times 4.3\cos(180-(29+"138.30203"))}$	
	$=1.95(\text{cm})$	A1
		[3]
ALT (b)	$180-(29+"138.30203")$ $(=12.69796.....)$	[M1]
	$\frac{\sin 12.7}{AC}=\frac{\sin 29}{4.3}$	
	$AC=0.2198\times\frac{4.3}{0.4848}$	[M1]
	$=1.95$ (cm)	[A1]
		[3]
		Total 7 marks

Type MS · Using the sine rule, including the two-answer case

Answer 6 of 14 · Question 4 part a · 4 marks

2025 JUN · Paper 01 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Using the sine rule, including the two-answer case

Answer 7 of 14 · Question 4 part b · 3 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

		For $t = \frac{}{4}$
(b)	M1	For setting $v = 0$ and attempting to solve, see general guidance for definition of a valid attempt to solve.
	A1	For one correct value of either T_1 or T_2
	A1	For both correct values
(c)	**Note:** **The question states 'Show your working clearly' No integration seen – award M0A0M0A0**	
	M1	For an attempt to integrate, see general guidance.
	A1	For a fully correct integration (constant of integration need not be shown).
	M1	For correct substitution of **their** T_1 and T_2 **into a changed expression AND subtracts their values correctly.** If these are not the correct values, then must see each limit correctly substituted into their expression at least once, If the correct values are used, but their final answer is not correct, explicit substitution must be seen for the award of this mark. **Note:** Mark may be implied by a correct answer given **following correct integration**
	A1	Correct answer **which must be a positive value.**

Type MS · Using the sine rule, including the two-answer case

Answer 7 of 14 · Question 4 part b · 3 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks
4 (a)	$\frac{\sin "A"}{5.9}=\frac{\sin 29}{4.3}$ oe	M1
	$(A=)\sin^{-1}\left(\frac{\sin 29}{4.3}\times 5.9\right)$ or $\sin^{-1}("0.66520.....")$ oe	dM1
	$A=41.7$	A1
	$A=180-"41.69796....."=138.3$	A1
		[4]
(b)	$180-(29+"138.30203")$ $(=12.69796.....)$	M1
	$(AC)^2=5.9^2+4.3^2-2\times 5.9\times 4.3\cos(180-(29+"138.30203"))$ or	M1
	$AC=\sqrt{5.9^2+4.3^2-2\times 5.9\times 4.3\cos(180-(29+"138.30203"))}$	
	$=1.95(\text{cm})$	A1
		[3]
ALT (b)	$180-(29+"138.30203")$ $(=12.69796.....)$	[M1]
	$\frac{\sin 12.7}{AC}=\frac{\sin 29}{4.3}$	
	$AC=0.2198\times\frac{4.3}{0.4848}$	[M1]
	$=1.95$ (cm)	[A1]
		[3]
		Total 7 marks

Type MS · Using the sine rule, including the two-answer case

Answer 7 of 14 · Question 4 part b · 3 marks

2025 JUN · Paper 01 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Using the sine rule, including the two-answer case

Answer 8 of 14 · Question 3 part whole · 7 marks

2022 JUN · Paper 01 · extract 1 of 8 · find part whole in this mark scheme extract

3	**Working in triangle *AED* – main scheme, ALT1 and ALT2 – first 3 marks:** $\frac{\sin E}{15}=\frac{\sin 55}{13}$ $E = 70.9...°$ or $109...°$ (awrt) $ADE = 180 - 55 -$ "109.1" (=15.9…°) or $180 - 55 -$ "70.9" (=54.0…°)	M1 A1 M1 (B1 on ePen)
Main Scheme	(Let h be the perpendicular height of triangle ADE) $\sin"15.9...°" = \frac{h}{13}$ **Allow use of "54.0"** $h = 3.57...$ or 10.5…..(awrt) Area $= \frac{1}{2}\times 15\times(10-'3.57...')$ **Do not allow use of *h* that has come from "54.0"** $= 48.2\ (\text{cm}^2)$(awrt)	M1 A1 ddM1 A1 (7)
ALT1	**First 3 marks as main scheme** $((AE)^2 =)13^2+15^2-2\times13\times15\cos("15.9....")$ **allow use of "54.0...." for "15.9...."** or $(AE =)\sqrt{13^2+15^2-2\times13\times15\cos("15.9....")}$ **allow use of "54.0...." for "15.9...."** $\frac{AE}{\sin"15.9"}=\frac{13}{\sin 55}$ **allow use of "54.0...." for "15.9...."** $(AE =)$awrt 4.35 or 4.36 **or** awrt12.8 or 12.9 if using using 54.0....rather than 15.9.... **FOR THE NEXT PART DO NOT ALLOW USE OF 54.0 or 12.8……..** $\frac{1}{2}\times"4.35...."\times10\times\sin35$ (Area $AEB = 12.4.....$) $\frac{1}{2}\times"4.35...."\times15\times\sin55$ *or* $\frac{1}{2}\times13\times15\times\sin"15.9...."$ (Area $AED = 26.7.....$) $\frac{1}{2}\times13\times10\times\sin"90-15.9"$ (Area $EDC = 62.5.....$) Area triangle $BCE = (10\times15) -$ (Area AEB + Area AED + Area EDC) awrt $48.2\ (\text{cm}^2)$	M1 A1 ddM1 A1 (7)
ALT2	**First 5 marks as ALT1** **DO NOT ALLOW USE OF 54.0 or 12.8……..for this 2 marks** $((EB)^2 =)10^2+"4.36"^2-2\times10\times"4.36"\cos(35)$ or $(EB =)\sqrt{10^2+"4.36"^2-2\times10\times"4.36"\cos(35)}$ $\frac{\sin EBA}{4.36.....}=\frac{\sin 35}{6.96....}\Rightarrow$ Angle $EBA = 21.2.....\Rightarrow$ Angle $EBC = 90-21.2..... = 68.7.....$ Area $= 0.5\times6.96\times15\times\sin(68.7.....)$ Area = awrt48.2	ddM1 A1 (7)

Type MS · Using the sine rule, including the two-answer case

Answer 8 of 14 · Question 3 part whole · 7 marks

2022 JUN · Paper 01 · extract 2 of 8 · find part whole in this mark scheme extract

ALT3	**Working in triangle *ABE***	M1
	or denoting side AE as y : $13^2 = 15^2 + y^2 - 2\times y\times 15\cos 55$	
	$\left(\Rightarrow 0 = y^2 - (17.2....)y + 56\right)$	
	$(y=)$awrt 4.35 *or* 4.36 or $(y=)$awrt 12.8	A1
	$c^2 = "4.36"^2 + 10^2 - 2\times "4.36"\times 10\cos 35$ **12.8 could also be used for 4.36**	M1
	$(c=$ awrt 6.89 or $7.38)$	
	$\dfrac{\sin 35}{"6.89" \text{or} "7.83"} = \dfrac{\sin ABE}{"4.36" \text{or} "12.8"}$	M1
	angle ABE = awrt 21.2 or 85.9 **and** c = awrt 6.89 or 7.38	A1
	Area $= \dfrac{1}{2}\times 15\times "6.89"\times \sin(90 - "21.2")$	ddM1
	awrt 48.2	A1
		Total 7 marks

Type MS · Using the sine rule, including the two-answer case

Answer 8 of 14 · Question 3 part whole · 7 marks

2022 JUN · Paper 01 · extract 3 of 8 · find part whole in this mark scheme extract

Part	Marks	Notes
(a)	B1	For $(8x+y)(6x+y)$ oe
(b)	M1	For their "$(8x+y)(6x+y)$" $-2(9x^2+15x^2)$ oe Neither of these expressions need to be in simplified form.
	M1	For correct expansion to give $48x^2+14xy+y^2$ - may be seen in part (a) or for $2(9x^2+15x^2)$ or $18x^2+30x^2$ or $48x^2$ written in any correct form – doesn't need to be simplified.
	A1 cso*	Obtains the given result with no errors seen.
	Note M0 M1 A0 is possible.	
ALT	M1	For an attempt to add or subtract separate components to give the area of the cross. Allow 1 omission or up to 2 errors. Examples are given in the MS, though these are not exhaustive.
	M1	All components present and correct and an attempt to simplify.
	A1 cso*	Obtains the given result with no errors seen.
	Note M0 M1 A0 is possible.	
(c)	M1	For solving simultaneously, using the correct equations, to obtain $48x^2+1080=3432$
	A1	For $x=7$ (must select the positive solution)
	M1	For substituting their value of x to obtain a 3TQ in terms of y only.
	M1	For an attempt to solve their 3TQ – see general guidance for minimally acceptable attempt.
	A1	For $y=10$ (must reject the other solution) final M1 A1 can be awarded for y = 10 with no working shown final M1 A0 can be awarded for y = 10 and y = -108.
ALT	M1	Correct rearrangement for x and correct substitution into $48x^2+14xy+y^2=3432$ reaching an equation of the form $ay^4+by^2+c=0$
	M1 (A1 on ePen)	For an attempt to solve their equation of the form $ay^4+by^2+c=0$ as a quadratic in y^2 (minimally acceptable attempt to solve, see general guidance) (A fully correct solution would give: $y^2=1164$ or $y^2=100$ or $y=\pm108$ or $y=\pm10$)
	A1 (M1 on ePen)	y = 10 (must clearly select this solution) 2nd M1 A1 can be awarded if no working and y = 10 2nd M1 A0 can be awarded if no working and $y=\pm108$ and $y=\pm10$ seen
	M1	For substituting their value for y and solving a linear equation to arrive at x =
	A1	$x=7$

Type MS · Using the sine rule, including the two-answer case

Answer 8 of 14 · Question 3 part whole · 7 marks

2022 JUN · Paper 01 · extract 4 of 8 · find part whole in this mark scheme extract

Question number	Scheme	Marks
3	**Working in triangle *AED* – main scheme, ALT1 and ALT2 – first 3 marks:** $\frac{\sin E}{15}=\frac{\sin 55}{13}$ $E = 70.9...°$ or $109...°$ (awrt) ADE = 180 – 55 – "109.1" (=15.9…°) or 180 – 55 – "70.9" (=54.0…°)	M1 A1 M1 (B1 on ePen)
Main Scheme	(Let h be the perpendicular height of triangle ADE) $\sin"15.9...°"=\frac{h}{13}$ **Allow use of "54.0"** $h = 3.57...$ or 10.5…..(awrt) Area = $\frac{1}{2}\times 15\times(10-'3.57...')$ **Do not allow use of *h* that has come from "54.0"** $= 48.2\ (\text{cm}^2)$(awrt)	M1 A1 ddM1 A1 (7)
ALT1	**First 3 marks as main scheme** $((AE)^2=)13^2+15^2-2\times13\times15\cos("15.9....")$ **allow use of "54.0...." for "15.9...."** or $(AE=)\sqrt{13^2+15^2-2\times13\times15\cos("15.9....")}$ **allow use of "54.0...." for "15.9...."** $\frac{AE}{\sin"15.9"}=\frac{13}{\sin 55}$ **allow use of "54.0...." for "15.9...."** $(AE=)$awrt 4.35 or 4.36 **or** awrt 12.8 or 12.9 if using using 54.0....rather than 15.9.... **FOR THE NEXT PART DO NOT ALLOW USE OF 54.0 or 12.8……..** $\frac{1}{2}\times"4.35...."\times10\times\sin 35$ (Area $AEB = 12.4.....$) $\frac{1}{2}\times"4.35...."\times15\times\sin 55$ *or* $\frac{1}{2}\times13\times15\times\sin"15.9...."$ (Area $AED = 26.7.....$) $\frac{1}{2}\times13\times10\times\sin"90-15.9"$ (Area $EDC = 62.5.....$) Area triangle $BCE = (10\times15)-$(Area AEB + Area AED + Area EDC) awrt 48.2 (cm^2)	M1 A1 ddM1 A1 (7)
ALT2	**First 5 marks as ALT1** **DO NOT ALLOW USE OF 54.0 or 12.8……..for this 2 marks** $((EB)^2=)10^2+"4.36"^2-2\times10\times"4.36"\cos(35)$ or $(EB=)\sqrt{10^2+"4.36"^2-2\times10\times"4.36"\cos(35)}$ $\frac{\sin EBA}{4.36.....}=\frac{\sin 35}{6.96....}\Rightarrow$ Angle $EBA = 21.2.....\Rightarrow$ Angle $EBC = 90-21.2..... = 68.7.....$ Area $= 0.5\times6.96\times15\times\sin(68.7.....)$ Area = awrt 48.2	ddM1 A1 (7)

Type MS · Using the sine rule, including the two-answer case

Answer 8 of 14 · Question 3 part whole · 7 marks

2022 JUN · Paper 01 · extract 5 of 8 · find part whole in this mark scheme extract

ALT3	**Working in triangle *ABE*** or denoting side AE as y : $13^2 = 15^2 + y^2 - 2\times y\times 15\cos 55$ $\left(\Rightarrow 0 = y^2 - (17.2....)y + 56\right)$	M1
	$(y =)$awrt 4.35 *or* 4.36 or $(y =)$awrt 12.8	A1
	$c^2 = "4.36"^2 + 10^2 - 2\times "4.36"\times 10\cos 35$ **12.8 could also be used for 4.36** $(c =$ awrt 6.89 or $7.38)$	M1
	$\dfrac{\sin 35}{"6.89" \text{or} "7.83"} = \dfrac{\sin ABE}{"4.36" \text{or} "12.8"}$	M1
	angle $ABE =$ awrt 21.2 or 85.9 **and** $c =$ awrt 6.89 or 7.38	A1
	Area $= \dfrac{1}{2}\times 15\times "6.89"\times \sin(90 - "21.2")$ awrt 48.2	ddM1 A1
		Total 7 marks
If the final answer given rounds to 48.2 and working is present, students may be given full marks if their intermediate values are not given to the degree of accuracy demanded in the mark scheme. Look carefully for answers written on the diagram and award marks.		

Mark	Notes for ALT3
M1	For a correct substitution into the cosine rule to find AE as shown
A1	For either y = awrt 4.35/4.36 or y = awrt 12.8
M1	Correct substitution into the cosine rule to find BE using their 4.36 or 12.8
M1	Correct substitution into the sine rule to find angle ABE
ddM1	For this mark, they must have used y = 4.35/4.36 throughout Correct substitution into the area of a triangle formula as shown
A1	awrt 48.2

Type MS · Using the sine rule, including the two-answer case

Answer 8 of 14 · Question 3 part whole · 7 marks

2022 JUN · Paper 01 · extract 6 of 8 · find part whole in this mark scheme extract

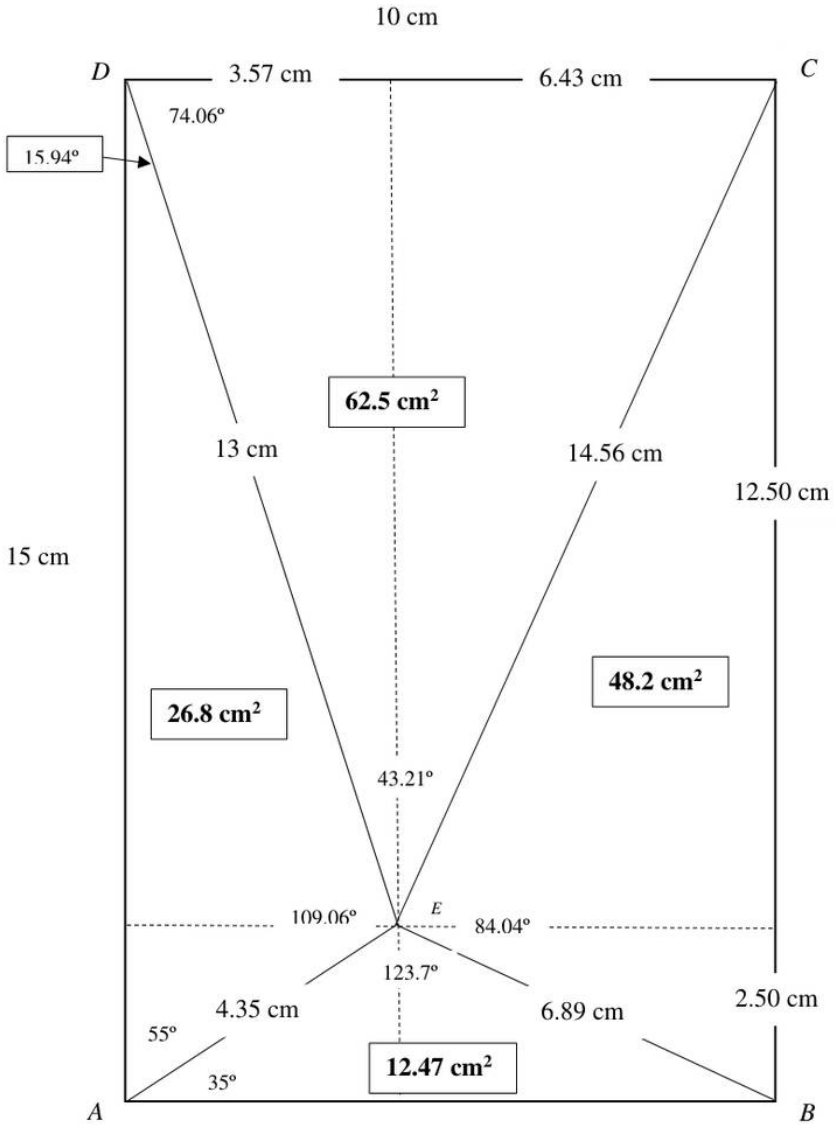
Marks	Notes
Working in triangle *AED* – main scheme, ALT1 and ALT2 – first 3 marks:	
M1	For correct use of the sine rule to find angle *AED*
A1	For either $AED = 70.9...°$ or $109...°$ awrt
M1 (B1 on ePen)	For *ADE* = (180 – 55 – their "109".1…)° or 180 – 55 – "70.9" (=54.0…°) It must be clear this is their angle *AED* if the angle is incorrect or for finding "74.1" – the angle which *h* makes with *DE* using "109.1" or "70.9"
Main scheme next 4 marks:	
M1	For $\sin"15.9"...° = \frac{h}{13}$ Use of their 15.9. Allow use of their 54.0…….. **Note, using the same triangle, this could also be presented as:** $\cos"74.1"...° = \frac{h}{13}$
A1	For $h = 3.57...$ or 10.5 (awrt)
ddM1	For area = $\frac{1}{2} \times 15 \times (10 - "3.57...")$ Dependent on both previous method marks – for this mark, candidates must have used angle *ADE* = (180 – 55 – their "109".1…="15.9")° ie **they must have used the obtuse angle found for angle *AED***
A1	For awrt 48.2 $\left(\text{cm}^2\right)$ (Units not required)
For all solutions: if the final answer given rounds to 48.2 and working is present, students may be given full marks if their intermediate values are not given to the degree of accuracy demanded in the mark scheme.	
ALT1	
M1	For correct substitution into cosine rule $13^2 + 15^2 - 2 \times 13 \times 15\cos("15.9....)"$ or $\sqrt{13^2 + 15^2 - 2 \times 13 \times 15\cos("15.9....")}$ Use of their 15.9….. Allow use of their 54.0……..
A1	For awrt 4.36
ddM1	For a **complete** method to find the area of triangle *BCE* as shown in the scheme – use of their values. Dependent on both previous method marks – for this mark, candidates must have used angle *ADE* = (180 – 55 – their "109".1…="15.9")° ie **they must have used the obtuse angle found for angle *AED***
A1	For awrt 48.2 $\left(\text{cm}^2\right)$ (Units not required)
ALT2 Final ddM1	For a **complete** method to find area of triangle *BCE* Finds *EB*, finds angle *EBC* Dependent on both previous method marks – for this mark, candidates must have used angle *ADE* = (180 – 55 – their "109".1…="15.9")° ie **they must have used the obtuse angle found for angle *AED***
A1	For awrt 48.2 $\left(\text{cm}^2\right)$ (Units not required)
General principles for marking this question with any other method seen: The first 5 marks will be allocated as the main scheme and/or the ALTS The final ddM1 mark will be awarded for a complete method, using their values **but MUST use the obtuse angle as stated in the question.** Final A1 for awrt 48.2	

USEFUL SKETCH

Type MS · Using the sine rule, including the two-answer case

Answer 8 of 14 · Question 3 part whole · 7 marks

2022 JUN · Paper 01 · extract 7 of 8 · find part whole in this mark scheme extract



Type MS · Using the sine rule, including the two-answer case

Answer 8 of 14 · Question 3 part whole · 7 marks

2022 JUN · Paper 01 · extract 8 of 8 · find part whole in this mark scheme extract

Question number	Scheme	Marks
	$a(1-r^4)$ a	

Type MS · Using the sine rule, including the two-answer case

Answer 9 of 14 · Question 9 part c · 4 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

9(a)	$\cos ADB = \dfrac{6^2 + x^2 - 12^2}{2\times 6x} = \dfrac{x^2 - 108}{12x}$	M1A1	(2)
(b)	$\cos BDC = \dfrac{6^2 + x^2 - 6^2}{2\times 6x} = \dfrac{x^2}{12x}$	B1	
	$ADB = 180 - BDC \Rightarrow -\dfrac{x^2}{12x} = \dfrac{x^2 - 108}{12x}$	M1A1	
	$2x^2 = 108 \Rightarrow x = 3\sqrt{6}$		
	$AC = 6\sqrt{6}$	A1	(4)
(c)	$\dfrac{\sin(\theta^\circ + \phi^\circ)}{2x} = \dfrac{\sin BCD}{12} \Rightarrow \dfrac{\sin(\theta^\circ + \phi^\circ)}{x} = \dfrac{\sin BCD}{6}$	M1A1	
	$\dfrac{\sin\phi^\circ}{x} = \dfrac{\sin BCD}{6}$	M1	
	$\therefore \sin\phi^\circ = \sin(\theta^\circ + \phi^\circ)$	A1	(4)
(d)	$\sin(\theta^\circ + \phi^\circ) = \sin\phi^\circ \Rightarrow (\theta + \phi) = 180 - \phi$ $(\text{or } \phi \text{ or } 360 + \phi \text{ (not possible)})$	M1	
	$\therefore \theta = 180 - 2\phi$	A1	(2)
			[12]

Part	Mark	Notes
(a)	M1	For using a correct cosine rule $\cos ADB = \dfrac{6^2 + x^2 - 12^2}{2\times 6x} = \dfrac{x^2 - 108}{12x}$ or $12^2 = x^2 + 6^2 - 2\times 6\times x\cos ADB$
	A1	Simplifies to $\cos ADB = \dfrac{x^2 - 108}{12x}$
(b)	B1	For the correct expression $\cos BDC = \dfrac{6^2 + x^2 - 6^2}{2\times 6x} = \dfrac{x^2}{12x}$
	M1	$\cos BDC = -\cos ADB$ and $\cos BDC = -\dfrac{x^2}{12x}$ so $-\dfrac{x^2}{12x} = \dfrac{x^2 - 108}{12x}$ and attempts to solve
	ALT 1 – uses triangles *BAD* and *BAC*	
	B1	For **both** of the following correct expressions for cos *BAD* and cos *BAC*: $\cos BAD = \dfrac{12^2 + x^2 - 6^2}{2\times 12\times x}$ and $\cos BAC = \dfrac{12^2 + (2x)^2 - 6^2}{2\times 12\times 2x}$
	M1	$\angle BAC = \angle BAD$ so equates their two expressions $\dfrac{12^2 + x^2 - 6^2}{2\times 12\times x} = \dfrac{12^2 + (2x)^2 - 6^2}{2\times 12\times 2x} \Rightarrow \dfrac{108 + x^2}{24x} = \dfrac{108 + 4x^2}{48x}$ and attempts to solve
	ALT 2 – uses triangles *BCD* and *BCA*	

Type MS · Using the sine rule, including the two-answer case

Answer 9 of 14 · Question 9 part c · 4 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

	B1	For **both** of the following correct expressions for cos BAD and cos BAC: $\cos BCD = \frac{6^2+x^2-6^2}{2\times6\times x}$ and $\cos BCA = \frac{6^2+(2x)^2-12^2}{2\times6\times2x}$
	M1	$\frac{6^2+x^2-6^2}{2\times6\times x} = \frac{6^2+(2x)^2-12^2}{2\times6\times2x} \Rightarrow x^2 = 2x^2-54$ and attempts to solve
	Final A marks for all three methods	
	A1	For the correct value of $x = 3\sqrt{6}$
	A1	For $AC = 6\sqrt{6}$
(c)	M1	Uses sine rule on triangle ABC: $\frac{\sin(\theta^\circ+\phi^\circ)}{2x} = \frac{\sin BCD}{12} \Rightarrow \frac{\sin(\theta^\circ+\phi^\circ)}{x} = \frac{\sin BCD}{6}$
	A1	Achieves the correct expression for $\sin(\theta^\circ+\phi^\circ) = \frac{x\sin BCD}{6}$
	M1	Uses sine rule on triangle BDC: $\frac{\sin\phi^\circ}{x} = \frac{\sin BCD}{6} \Rightarrow \left(\sin\phi^\circ = \frac{x\sin BCD}{6}\right)$
	A1	Shows that $\sin\phi^\circ = \sin(\theta^\circ+\phi^\circ)$ with no errors
	ALT 1 – Uses exact values for the trigonometric ratios and the expansion for sin $(A + B)$	
	M1	Finds $\cos\theta = \frac{7}{8} \Rightarrow \sin\theta = \frac{\sqrt{15}}{8}$ **or** $\cos\phi = \frac{1}{4} \Rightarrow \sin\phi = \frac{\sqrt{15}}{4}$ Accept $\theta = 28.95...^\circ$ **or** $\phi = 75.52...^\circ \Rightarrow \sin\phi = 0.968...$
	A1	Finds $\cos\theta = \frac{7}{8} \Rightarrow \sin\theta = \frac{\sqrt{15}}{8}$ **and** $\cos\phi = \frac{1}{4} \Rightarrow \sin\phi = \frac{\sqrt{15}}{4}$ Accept $\theta = 28.95...^\circ$ **and** $\phi = 75.52...^\circ \Rightarrow \sin\phi = 0.968...$
	M1	Expands $\sin(\theta+\phi) = \frac{\sqrt{15}}{8}\times\frac{1}{4} + \frac{\sqrt{15}}{4}\times\frac{7}{8} = \left(\frac{\sqrt{15}}{4}\right)$ Or $\sin(\theta+\phi)^\circ = \sin(28.95^\circ)\cos(75.52^\circ)+\sin(75.52^\circ)\cos(28.95^\circ) = 0.968... = \sin\phi^\circ$
	A1	Shows that $\sin(\theta+\phi) = \frac{\sqrt{15}}{4}$ and $\sin\phi = \frac{\sqrt{15}}{4}$ so $\sin(\theta+\phi) = \sin\phi$ with no errors. If they use approximate values for sin θ and ϕ withhold this final mark so A0
	ALT 2 – Uses $\angle BCD = 52.2...^\circ$	
	M1	Finds $\angle BCD = 52.2...^\circ$ using cosine rule and applies sine rule on triangle ABC $\frac{\sin(\theta^\circ+\phi^\circ)}{6\sqrt{6}} = \frac{\sin 52.2^\circ}{12}$
	A1	Shows that $\sin(\theta^\circ+\phi^\circ) = 0.968...$
	M1	Uses sine rule on triangle BD: $\frac{\sin\phi^\circ}{3\sqrt{6}} = \frac{\sin 52.2^\circ}{6} \Rightarrow \sin\phi^\circ = 0.968... = \sin(\theta^\circ+\phi^\circ)$
	A1	If they use an approximate value for angle BCD withhold this final mark so A0
(d)	M1	For writing $\sin\phi^\circ = \sin(180-\phi)^\circ \Rightarrow \theta^\circ+\phi^\circ = 180^\circ-\phi^\circ$
	A1	For rearranging $\theta^\circ+\phi^\circ = 180^\circ-\phi^\circ$ to achieve $\therefore \theta = 180-2\phi$ This is a show question and there must be no errors here.

Type MS · Using the sine rule, including the two-answer case

Answer 9 of 14 · Question 9 part c · 4 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Using the sine rule, including the two-answer case

Answer 10 of 14 · Question 9 part d · 2 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part d in this mark scheme extract

9(a)	$\cos ADB = \dfrac{6^2 + x^2 - 12^2}{2\times 6x} = \dfrac{x^2 - 108}{12x}$	M1A1	(2)
(b)	$\cos BDC = \dfrac{6^2 + x^2 - 6^2}{2\times 6x} = \dfrac{x^2}{12x}$	B1	
	$ADB = 180 - BDC \Rightarrow -\dfrac{x^2}{12x} = \dfrac{x^2 - 108}{12x}$	M1A1	
	$2x^2 = 108 \Rightarrow x = 3\sqrt{6}$		
	$AC = 6\sqrt{6}$	A1	(4)
(c)	$\dfrac{\sin(\theta^\circ + \phi^\circ)}{2x} = \dfrac{\sin BCD}{12} \Rightarrow \dfrac{\sin(\theta^\circ + \phi^\circ)}{x} = \dfrac{\sin BCD}{6}$	M1A1	
	$\dfrac{\sin\phi^\circ}{x} = \dfrac{\sin BCD}{6}$	M1	
	$\therefore \sin\phi^\circ = \sin(\theta^\circ + \phi^\circ)$	A1	(4)
(d)	$\sin(\theta^\circ + \phi^\circ) = \sin\phi^\circ \Rightarrow (\theta + \phi) = 180 - \phi$ $(\text{or } \phi \text{ or } 360 + \phi \text{ (not possible)})$	M1	
	$\therefore \theta = 180 - 2\phi$	A1	(2)
			[12]

Part	Mark	Notes
(a)	M1	For using a correct cosine rule $\cos ADB = \dfrac{6^2 + x^2 - 12^2}{2\times 6x} = \dfrac{x^2 - 108}{12x}$ or $12^2 = x^2 + 6^2 - 2\times 6\times x\cos ADB$
	A1	Simplifies to $\cos ADB = \dfrac{x^2 - 108}{12x}$
(b)	B1	For the correct expression $\cos BDC = \dfrac{6^2 + x^2 - 6^2}{2\times 6x} = \dfrac{x^2}{12x}$
	M1	$\cos BDC = -\cos ADB$ and $\cos BDC = -\dfrac{x^2}{12x}$ so $-\dfrac{x^2}{12x} = \dfrac{x^2 - 108}{12x}$ and attempts to solve
	ALT 1 – uses triangles *BAD* and *BAC*	
	B1	For **both** of the following correct expressions for cos *BAD* and cos *BAC*: $\cos BAD = \dfrac{12^2 + x^2 - 6^2}{2\times 12\times x}$ and $\cos BAC = \dfrac{12^2 + (2x)^2 - 6^2}{2\times 12\times 2x}$
	M1	$\angle BAC = \angle BAD$ so equates their two expressions $\dfrac{12^2 + x^2 - 6^2}{2\times 12\times x} = \dfrac{12^2 + (2x)^2 - 6^2}{2\times 12\times 2x} \Rightarrow \dfrac{108 + x^2}{24x} = \dfrac{108 + 4x^2}{48x}$ and attempts to solve
	ALT 2 – uses triangles *BCD* and *BCA*	

Type MS · Using the sine rule, including the two-answer case

Answer 10 of 14 · Question 9 part d · 2 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part d in this mark scheme extract

	B1	For **both** of the following correct expressions for cos BAD and cos BAC: $\cos BCD = \frac{6^2 + x^2 - 6^2}{2\times 6\times x}$ and $\cos BCA = \frac{6^2 + (2x)^2 - 12^2}{2\times 6\times 2x}$
	M1	$\frac{6^2 + x^2 - 6^2}{2\times 6\times x} = \frac{6^2 + (2x)^2 - 12^2}{2\times 6\times 2x} \Rightarrow x^2 = 2x^2 - 54$ and attempts to solve
	Final A marks for all three methods	
	A1	For the correct value of $x = 3\sqrt{6}$
	A1	For $AC = 6\sqrt{6}$
(c)	M1	Uses sine rule on triangle ABC: $\frac{\sin(\theta^\circ + \phi^\circ)}{2x} = \frac{\sin BCD}{12} \Rightarrow \frac{\sin(\theta^\circ + \phi^\circ)}{x} = \frac{\sin BCD}{6}$
	A1	Achieves the correct expression for $\sin(\theta^\circ + \phi^\circ) = \frac{x \sin BCD}{6}$
	M1	Uses sine rule on triangle BDC: $\frac{\sin\phi^\circ}{x} = \frac{\sin BCD}{6} \Rightarrow \left(\sin\phi^\circ = \frac{x\sin BCD}{6}\right)$
	A1	Shows that $\sin\phi^\circ = \sin(\theta^\circ + \phi^\circ)$ with no errors
	ALT 1 – Uses exact values for the trigonometric ratios and the expansion for sin $(A + B)$	
	M1	Finds $\cos\theta = \frac{7}{8} \Rightarrow \sin\theta = \frac{\sqrt{15}}{8}$ **or** $\cos\phi = \frac{1}{4} \Rightarrow \sin\phi = \frac{\sqrt{15}}{4}$ Accept $\theta = 28.95...^\circ$ **or** $\phi = 75.52...^\circ \Rightarrow \sin\phi = 0.968...$
	A1	Finds $\cos\theta = \frac{7}{8} \Rightarrow \sin\theta = \frac{\sqrt{15}}{8}$ **and** $\cos\phi = \frac{1}{4} \Rightarrow \sin\phi = \frac{\sqrt{15}}{4}$ Accept $\theta = 28.95...^\circ$ **and** $\phi = 75.52...^\circ \Rightarrow \sin\phi = 0.968...$
	M1	Expands $\sin(\theta + \phi) = \frac{\sqrt{15}}{8}\times\frac{1}{4} + \frac{\sqrt{15}}{4}\times\frac{7}{8} = \left(\frac{\sqrt{15}}{4}\right)$ Or $\sin(\theta + \phi)^\circ = \sin(28.95^\circ)\cos(75.52^\circ) + \sin(75.52^\circ)\cos(28.95^\circ) = 0.968... = \sin\phi^\circ$
	A1	Shows that $\sin(\theta + \phi) = \frac{\sqrt{15}}{4}$ and $\sin\phi = \frac{\sqrt{15}}{4}$ so $\sin(\theta + \phi) = \sin\phi$ with no errors. If they use approximate values for sin θ and ϕ withhold this final mark so A0
	ALT 2 – Uses $\angle BCD = 52.2...^\circ$	
	M1	Finds $\angle BCD = 52.2...^\circ$ using cosine rule and applies sine rule on triangle ABC $\frac{\sin(\theta^\circ + \phi^\circ)}{6\sqrt{6}} = \frac{\sin 52.2^\circ}{12}$
	A1	Shows that $\sin(\theta^\circ + \phi^\circ) = 0.968...$
	M1	Uses sine rule on triangle BD: $\frac{\sin\phi^\circ}{3\sqrt{6}} = \frac{\sin 52.2^\circ}{6} \Rightarrow \sin\phi^\circ = 0.968... = \sin(\theta^\circ + \phi^\circ)$
	A1	If they use an approximate value for angle BCD withhold this final mark so A0
(d)	M1	For writing $\sin\phi^\circ = \sin(180 - \phi)^\circ \Rightarrow \theta^\circ + \phi^\circ = 180^\circ - \phi^\circ$
	A1	For rearranging $\theta^\circ + \phi^\circ = 180^\circ - \phi^\circ$ to achieve $\therefore \theta = 180 - 2\phi$ This is a show question and there must be no errors here.

Type MS · Using the sine rule, including the two-answer case

Answer 10 of 14 · Question 9 part d · 2 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part d in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Using the sine rule, including the two-answer case

Answer 11 of 14 · Question 6 part a.i · 0 marks

2022 JAN · Paper 01 · extract 1 of 4 · find part a.i in this mark scheme extract

6(a)(i)	$\frac{\sin 30^\circ}{x} = \frac{\sin ACB}{x+3}$ $\sin\theta^\circ = \frac{x+3}{2x}$ *	M1 A1 cso
(ii)	$\cos^2\theta^\circ = 1-\left(\frac{x+3}{2x}\right)^2$ $\cos^2\theta^\circ = \frac{(2x)^2-(x+3)^2}{(2x)^2}$ $\cos\theta^\circ = \frac{\sqrt{3x^2-6x-9}}{2x}$ *	M1 M1 A1 [5]
Alt – use of right-angled triangle with Pythagoras' theorem		
	Adjacent $= \sqrt{(2x)^2-(x+3)^2}$ $\cos\theta = \frac{\sqrt{(2x)^2-(x+3)^2}}{2x}$ $\cos\theta^\circ = \frac{\sqrt{3x^2-6x-9}}{2x}$ *	[M1 M1 A1]
(b)	$\frac{\angle BAC}{30^\circ} = \frac{7}{2} \Rightarrow \angle BAC = 105^\circ$ $\theta = 180-30-105 = 45$ $\cos 45^o = \frac{\sqrt{2}}{2} = \frac{\sqrt{3x^2-6x-9}}{2x} \Rightarrow 2x^2 = 3x^2-6x-9 \Rightarrow x^2-6x-9=0$ $x^2-6x-9=0 \Rightarrow (x-3)^2-18=0 \Rightarrow x=\ldots$ $x = 3+3\sqrt{2}$	B1 B1 M1 M1 A1 [5]
Alt – last three marks		
	$\sin\theta^\circ = \frac{x+3}{2x} = \frac{\sqrt{2}}{2} \Rightarrow x+3 = \sqrt{2}x$ $x+3 = \sqrt{2}x \Rightarrow x\left(\sqrt{2}-1\right) = 3 \Rightarrow x = \frac{3}{\sqrt{2}-1}$ $x = 3+3\sqrt{2}$	[M1 M1 A1]
		Total 10 marks

Type MS · Using the sine rule, including the two-answer case

Answer 11 of 14 · Question 6 part a.i · 0 marks

2022 JAN · Paper 01 · extract 2 of 4 · find part a.i in this mark scheme extract

Part	Marks	Scheme
(a) (i)	M1	For using a correct sine rule to give, $\frac{\sin 30^\circ}{x} = \frac{\sin ACB}{x+3}$
	A1 cso	For correctly obtaining the expression for $\sin\theta$ $\sin\theta^\circ = \frac{x+3}{2x}$ *
(ii)	M1	For using the Pythagorean identity $\cos^2\theta^\circ = 1-\left(\frac{x+3}{2x}\right)^2$
	M1	For simplifying to form a single fraction $\cos^2\theta^\circ = \frac{(2x)^2-(x+3)^2}{(2x)^2}$
	A1 cso	For simplifying to achieve the given expression, $\cos\theta^\circ = \frac{\sqrt{3x^2-6x-9}}{2x}$ * Note this is a show question
Alt – use of right-angled triangle with Pythagoras' theorem		
	M1	For use of a right-angled triangle with Pythagoras' theorem to determine the adjacent Adjacent $= \sqrt{(2x)^2-(x+3)^2}$
	M1	For use of cosine ratio $\cos\theta = \frac{\sqrt{(2x)^2-(x+3)^2}}{2x}$
	A1 cso	For simplifying to achieve the given expression, $\cos\theta^\circ = \frac{\sqrt{3x^2-6x-9}}{2x}$ * Note this is a show question
(b)	B1	For finding the size of $\angle BAC$ $\frac{\angle BAC}{30^\circ} = \frac{7}{2} \Rightarrow \angle BAC = 105^\circ$
	B1	For finding the value of $\theta = 180-30-105 = 45$
	M1	For substituting the value of $\angle ABC$ into the given expression for $\cos\theta$ and forming a 3TQ, condone arithmetic errors in rearrangement. $\cos 45^o = \frac{\sqrt{2}}{2} = \frac{\sqrt{3x^2-6x-9}}{2x} \Rightarrow 2x^2 = 3x^2-6x-9 \Rightarrow x^2-6x-9=0$ Allow for use of their 45° but this must come from an attempt at working with the ratio. Do not allow if their 45° is 30°.
	M1	For an attempt to solve their 3TQ by any valid method (see general guidance) $x^2-6x-9=0 \Rightarrow (x-3)^2-18=0 \Rightarrow x=...$
	A1 cao	For the correct value of x in the correct form $x = 3+3\sqrt{2}$ Allow $a = 3, b = 2$
Alternative method		
	M1	For substituting the value of $\angle ABC$ into the given expression for $\sin\theta$ and forming a linear equation $\sin\theta^\circ = \frac{x+3}{2x} = \frac{\sqrt{2}}{2} \Rightarrow x+3 = \sqrt{2}x$

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Using the sine rule, including the two-answer case

Answer 11 of 14 · Question 6 part a.i · 0 marks

2022 JAN · Paper 01 · extract 3 of 4 · find part a.i in this mark scheme extract

	M1	For an attempt to solve their linear equation $x+3=\sqrt{2}x \Rightarrow x\left(\sqrt{2}-1\right)=3 \Rightarrow x=\dfrac{3}{\sqrt{2}-1}$
	A1 **cao**	For the correct value of x in the correct form $x=3+3\sqrt{2}$ Allow $a = 3, b = 2$

18

Type MS · Using the sine rule, including the two-answer case

Answer 11 of 14 · Question 6 part a.i · 0 marks

2022 JAN · Paper 01 · extract 4 of 4 · find part a.i in this mark scheme extract

Question	Scheme	Marks

Type MS · Using the sine rule, including the two-answer case

Answer 12 of 14 · Question 6 part b · 5 marks

2022 JAN · Paper 01 · extract 1 of 4 · find part b in this mark scheme extract

6(a)(i)	$\frac{\sin 30^\circ}{x} = \frac{\sin ACB}{x+3}$ $\sin\theta^\circ = \frac{x+3}{2x}$ *	M1 A1 cso
(ii)	$\cos^2\theta^\circ = 1-\left(\frac{x+3}{2x}\right)^2$ $\cos^2\theta^\circ = \frac{(2x)^2-(x+3)^2}{(2x)^2}$ $\cos\theta^\circ = \frac{\sqrt{3x^2-6x-9}}{2x}$ *	M1 M1 A1 [5]
Alt – use of right-angled triangle with Pythagoras' theorem		
	Adjacent $= \sqrt{(2x)^2-(x+3)^2}$ $\cos\theta = \frac{\sqrt{(2x)^2-(x+3)^2}}{2x}$ $\cos\theta^\circ = \frac{\sqrt{3x^2-6x-9}}{2x}$ *	[M1 M1 A1]
(b)	$\frac{\angle BAC}{30^\circ} = \frac{7}{2} \Rightarrow \angle BAC = 105^\circ$ $\theta = 180-30-105 = 45$ $\cos 45^o = \frac{\sqrt{2}}{2} = \frac{\sqrt{3x^2-6x-9}}{2x} \Rightarrow 2x^2 = 3x^2-6x-9 \Rightarrow x^2-6x-9=0$ $x^2-6x-9=0 \Rightarrow (x-3)^2-18=0 \Rightarrow x=\ldots$ $x = 3+3\sqrt{2}$	B1 B1 M1 M1 A1 [5]
Alt – last three marks		
	$\sin\theta^\circ = \frac{x+3}{2x} = \frac{\sqrt{2}}{2} \Rightarrow x+3 = \sqrt{2}x$ $x+3 = \sqrt{2}x \Rightarrow x\left(\sqrt{2}-1\right) = 3 \Rightarrow x = \frac{3}{\sqrt{2}-1}$ $x = 3+3\sqrt{2}$	[M1 M1 A1]
		Total 10 marks

Type MS · Using the sine rule, including the two-answer case

Answer 12 of 14 · Question 6 part b · 5 marks

2022 JAN · Paper 01 · extract 2 of 4 · find part b in this mark scheme extract

Part	Marks	Scheme
(a) (i)	M1	For using a correct sine rule to give, $\frac{\sin 30^\circ}{x} = \frac{\sin ACB}{x+3}$
	A1 cso	For correctly obtaining the expression for $\sin\theta$ $\sin\theta^\circ = \frac{x+3}{2x}$ *
(ii)	M1	For using the Pythagorean identity $\cos^2\theta^\circ = 1-\left(\frac{x+3}{2x}\right)^2$
	M1	For simplifying to form a single fraction $\cos^2\theta^\circ = \frac{(2x)^2-(x+3)^2}{(2x)^2}$
	A1 cso	For simplifying to achieve the given expression, $\cos\theta^\circ = \frac{\sqrt{3x^2-6x-9}}{2x}$ * Note this is a show question
Alt – use of right-angled triangle with Pythagoras' theorem		
	M1	For use of a right-angled triangle with Pythagoras' theorem to determine the adjacent Adjacent $= \sqrt{(2x)^2-(x+3)^2}$
	M1	For use of cosine ratio $\cos\theta = \frac{\sqrt{(2x)^2-(x+3)^2}}{2x}$
	A1 cso	For simplifying to achieve the given expression, $\cos\theta^\circ = \frac{\sqrt{3x^2-6x-9}}{2x}$ * Note this is a show question
(b)	B1	For finding the size of $\angle BAC$ $\frac{\angle BAC}{30^\circ} = \frac{7}{2} \Rightarrow \angle BAC = 105^\circ$
	B1	For finding the value of $\theta = 180-30-105 = 45$
	M1	For substituting the value of $\angle ABC$ into the given expression for $\cos\theta$ and forming a 3TQ, condone arithmetic errors in rearrangement. $\cos 45^o = \frac{\sqrt{2}}{2} = \frac{\sqrt{3x^2-6x-9}}{2x} \Rightarrow 2x^2 = 3x^2-6x-9 \Rightarrow x^2-6x-9=0$ Allow for use of their 45° but this must come from an attempt at working with the ratio. Do not allow if their 45° is 30°.
	M1	For an attempt to solve their 3TQ by any valid method (see general guidance) $x^2-6x-9=0 \Rightarrow (x-3)^2-18=0 \Rightarrow x=...$
	A1 cao	For the correct value of x in the correct form $x = 3+3\sqrt{2}$ Allow $a = 3, b = 2$
Alternative method		
	M1	For substituting the value of $\angle ABC$ into the given expression for $\sin\theta$ and forming a linear equation $\sin\theta^\circ = \frac{x+3}{2x} = \frac{\sqrt{2}}{2} \Rightarrow x+3 = \sqrt{2}x$

Type MS · Using the sine rule, including the two-answer case

Answer 12 of 14 · Question 6 part b · 5 marks

2022 JAN · Paper 01 · extract 3 of 4 · find part b in this mark scheme extract

	M1	For an attempt to solve their linear equation $x+3=\sqrt{2}x \Rightarrow x\left(\sqrt{2}-1\right)=3 \Rightarrow x=\dfrac{3}{\sqrt{2}-1}$
	A1 **cao**	For the correct value of x in the correct form $x=3+3\sqrt{2}$ Allow $a = 3, b = 2$

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Type MS · Using the sine rule, including the two-answer case

Answer 12 of 14 · Question 6 part b · 5 marks

2022 JAN · Paper 01 · extract 4 of 4 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Using the sine rule, including the two-answer case

Answer 13 of 14 · Question 6 part b · 3 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

6 (a)	**Method 1** $\left(\frac{a+\sqrt{5}}{\sqrt{5}-2}\times\frac{\sqrt{5}+2}{\sqrt{5}+2}\right)=\frac{\sqrt{5}a+2a+5+2\sqrt{5}}{(1)}$ $=\left[2a+5+(2+a)\sqrt{5}\right]$ $a=3$	**Method 2** $((11+5\sqrt{5})(\sqrt{5}-2))$ $11\sqrt{5}-22+25-10\sqrt{5}$ $a=3$	M1 A1 [2]
(b)	(angle PQR =) 45° $\frac{x+3}{\sin 45^\circ}=\frac{x}{\sin 30^\circ}\Rightarrow\frac{x+3}{\frac{\sqrt{2}}{2}}=\frac{x}{\frac{1}{2}}\Rightarrow x+3=\sqrt{2}x$ $\Rightarrow 3=x\left(\sqrt{2}-1\right)\Rightarrow x=\frac{3}{\sqrt{2}-1}\times\frac{\sqrt{2}+1}{\sqrt{2}+1}\Rightarrow x=\ldots$ $x=3+3\sqrt{2}$ **OR** $\frac{x+3}{\sin 45^\circ}=\frac{x}{\sin 30^\circ}\Rightarrow\frac{x+3}{\frac{\sqrt{2}}{2}}=\frac{x}{\frac{1}{2}}\Rightarrow\sqrt{2}x+3\sqrt{2}=2x$ $x\left(2-\sqrt{2}\right)=3\sqrt{2}\Rightarrow x=\frac{3\sqrt{2}}{2-\sqrt{2}}\times\frac{2+\sqrt{2}}{2+\sqrt{2}}\Rightarrow x=$ $x=3+3\sqrt{2}$		M1 M1 A1* cso [3] [M1 M1 A1]
(c)	$\frac{1}{2}\times(3+3\sqrt{2})\times(6+3\sqrt{2})\times\sin(180-30-45)^\circ$ $\frac{1}{2}\times(3+3\sqrt{2})\times(6+3\sqrt{2})\times\left(\frac{\sqrt{6}+\sqrt{2}}{4}\right)$ $\frac{9}{8}\left(4\sqrt{6}+4\sqrt{2}+3\sqrt{12}+6\right)$ or $\frac{1}{8}\left(36\sqrt{6}+36\sqrt{2}+54\sqrt{12}+54\right)$ oe with two out of four terms correct. $\frac{9}{8}\left(4\sqrt{6}+4\sqrt{2}+6\sqrt{3}+6\right)$		M1 dM1 A1 cao [3]
			Total 8 marks

Type MS · Using the sine rule, including the two-answer case

Answer 13 of 14 · Question 6 part b · 3 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
Throughout this question allow missing degree signs.		
(a)	M1	Any three out of four terms correct in the numerator of method 1 or in the expansion of method 2. If they use method 1 we must see the expansion come from multiplying numerator and denominator by $\sqrt{5}+2$
	A1	Cao If the answer $a = 3$ is stated without sufficient working seen award M0A0.
(b)	M1	For using a correct sine rule, either way around, using the exact values of sin 45 and sin 30, and attempting to simplify to $kx+3=\sqrt{2}x$ where k is a constant. Allow $\frac{kx}{2}+\frac{3}{2}=\frac{\sqrt{2}x}{2}$ for this mark
	M1	For factorising their $x+3=\sqrt{2}x$ [which must have come from using sine rule] to obtain $3=x\left(\sqrt{2}-k\right)$, rearranging and multiplying numerator and denominator by $\sqrt{2}+k$ This is a given answer – this step must be seen explicitly
	A1*	cso, no errors or emissions.
(c)	M1	Follow general guidance, if area formula is quoted, allow one slip in substitution. If formula not quoted, must be a fully correct substitution. For this mark only allow the angle to be $(180-30-45)^\circ$ If they work in terms of x and obtain $\frac{1}{2}\times x\times(x+3)\times\sin 105^\circ$ then do not award the first M mark until they have substituted the given value of $3+3\sqrt{2}$ for x.
	dM1	Replaces $\sin 105^\circ$ with the given value and attempts to multiply out. The mark can be awarded for any two terms correct, four terms must be present.
	A1cao	Must be in the given form with the terms in any order.

Type MS · Using the sine rule, including the two-answer case

Answer 13 of 14 · Question 6 part b · 3 marks

2022 JUN · Paper 01 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Using the sine rule, including the two-answer case

Answer 14 of 14 · Question 9 part b · 5 marks

2024 JUN · Paper 02 · extract 1 of 4 · find part b in this mark scheme extract

9(a)	$\sin 75^\circ = \sin\left(45^\circ + 30^\circ\right) = \sin 45^\circ \cos 30^\circ + \sin 30^\circ \cos 45^\circ$ $= \frac{\sqrt{2}}{2} \times \frac{\sqrt{3}}{2} + \frac{1}{2} \times \frac{\sqrt{2}}{2} = \frac{\sqrt{6}+\sqrt{2}}{4}$ *	M1 M1A1 cso
	ALT 1 $\sin 75^\circ = \sin\left(120^\circ - 45^\circ\right) = \sin 120^\circ \cos 45^\circ - \sin 45^\circ \cos 120^\circ$ $= \frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} \times \left(-\frac{1}{2}\right) = \frac{\sqrt{6}+\sqrt{2}}{4}$ *	[3] [M1 M1A1]
	ALT 2 $\sin 75^\circ = \sin\left(60^\circ + 15^\circ\right) = \sin 60^\circ \cos 15^\circ + \sin 15^\circ \cos 60^\circ$ $= \frac{\sqrt{3}}{2} \times \left(\frac{\sqrt{6}+\sqrt{2}}{4}\right) + \left(\frac{\sqrt{6}-\sqrt{2}}{4}\right) \times \frac{1}{2}$ $= \frac{3\sqrt{2}+\sqrt{6}}{8} + \frac{\sqrt{6}-\sqrt{2}}{8} = \frac{\sqrt{6}+\sqrt{2}}{4}$	[M1 M1 A1]
(b)	$\left[\angle AFC = 180^\circ - 60^\circ - 45^\circ = 75^\circ\right]$ $\frac{24}{\sin 75^\circ} = \frac{BF}{\sin 60^\circ}$ $\Rightarrow \frac{24}{\sin 75^\circ} = \frac{BF}{\sin 60^\circ} \Rightarrow \frac{24}{\frac{\sqrt{6}+\sqrt{2}}{4}} = \frac{BF}{\frac{\sqrt{3}}{2}}, \Rightarrow BF = \frac{48\sqrt{3}}{\sqrt{6}+\sqrt{2}}$ $BF = \frac{48\sqrt{3}}{\left(\sqrt{6}+\sqrt{2}\right)} \times \frac{\left(\sqrt{6}-\sqrt{2}\right)}{\left(\sqrt{6}-\sqrt{2}\right)} = \frac{48\sqrt{3}\sqrt{6} - 48\sqrt{6}}{6-2} = 12\left(3\sqrt{2} - \sqrt{6}\right)$ *	M1,M1A1 M1A1 cso [5]
(c)	Let the vertical point from F to the line AB be X and the corresponding point on CD be Y. The required angle is EXY $FX = 12\left(3\sqrt{2} - \sqrt{6}\right)\sin 45^\circ = 36 - 12\sqrt{3} = (15.2153...)$ OR Area $ABC = \frac{1}{2} \times 24 \times FX = \frac{1}{2} \times 24 \times 12\left(3\sqrt{2} - \sqrt{6}\right)\sin 45$ $\Rightarrow FX = 12\left(3\sqrt{2} - \sqrt{6}\right)\sin 45^\circ = 36 - 12\sqrt{3} = (15.2153...)$ $EF = \frac{36 - 12\sqrt{3}}{\tan 65^\circ} = 7.09505.... \approx 7.1$ (cm)	M1 M1A1 [3]
(d)	Identifies the required angle FCX $FC = \sqrt{\left[12\left(3\sqrt{2} - \sqrt{6}\right)\right]^2 + '7.09505'^2} = 22.65735.....$ $\angle FCX = \sin^{-1}\left(\frac{15.2153}{'22.6573'}\right) = 42.186...^\circ \approx 42.2^\circ$	B1 M1 M1A1 [4]
		Total 15 marks

Type MS · Using the sine rule, including the two-answer case

Answer 14 of 14 · Question 9 part b · 5 marks

2024 JUN · Paper 02 · extract 2 of 4 · find part b in this mark scheme extract

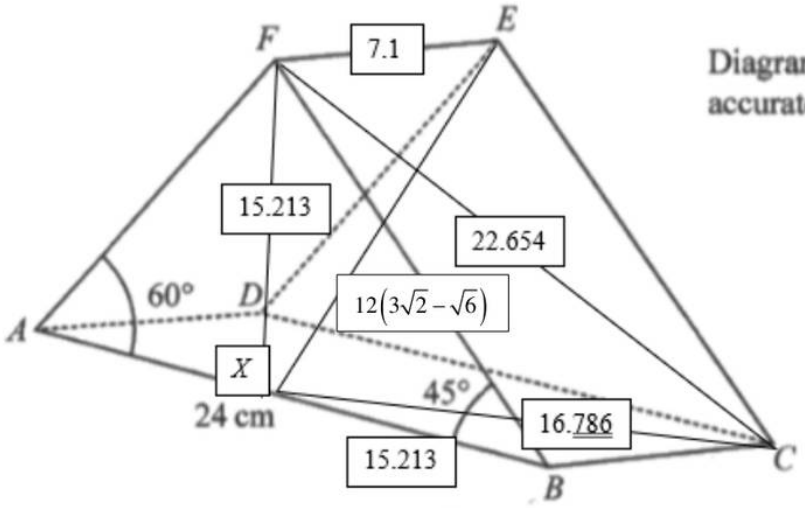
Part	Mark	Notes
(a)	M1	For the correct use of the addition formula for sine with $\sin(45^\circ+30^\circ)$ Accept also $\sin(120^\circ-45^\circ)$ and $\sin(60^\circ+15^\circ)$ **but not** $\sin(90^\circ-15^\circ)$ or $\sin(180^\circ-105^\circ)$
	M1	For applying the exact values of these sines and cosines and for evaluating the expression.. This step must be explicitly seen.
	A1 cso	For the correct final answer with no errors seen
(b)	M1	For the correct use of sine rule using the exact value for sin 75° and sin 60°
	M1	For rearranging to obtain $BF=\frac{k\sqrt{3}}{\sqrt{6}+\sqrt{2}}$ or $\frac{l\sqrt{3}}{2\sqrt{6}+2\sqrt{2}}$ Where k and l are integers
	A1	For the correct **exact** BF which need not be simplified. $BF=\frac{48\sqrt{3}}{\sqrt{6}+\sqrt{2}}$ or $\frac{96\sqrt{3}}{2\sqrt{6}+2\sqrt{2}}$ or even $\frac{12\sqrt{3}\times4}{\sqrt{6}+\sqrt{2}}$ but not for example, $\frac{\frac{12\sqrt{3}}{\sqrt{6}+\sqrt{2}}}{4}$ unless recovery is seen later
	M1	For multiplying numerator **and** denominator by their conjugate which must of the form e.g $\sqrt{A}\pm\sqrt{B}$ **AND** simplifying to the required form. We need to see at least one intermediate line of working before the final answer here.
	A1 cso	The answer as shown with no errors or omissions seen. This is a given result.
(c)	M1	For finding the length FX using any appropriate trigonometry.
	M1	For using any appropriate trigonometry to find length EF [or XY]
	A1	For awrt 7.1 (cm)
(d)	B1	For identifying the correct angle required.
	M1	For using Pythagoras theorem to find FC $FC=\sqrt{21.51781^2+7.09505^2}=22.65735.....$ Or For using Pythagoras theorem to find CX $FC=\sqrt{15.213^2+7.095^2}=16.786...$ [Note: $BX=15.213...$ and $CX=16.786\ldots$]
	M1	For using any appropriate trigonometry to find angle FCX. For example; $\angle FCX=\tan^{-1}\left(\frac{15.2153}{16.786}\right)=42.186...^\circ\approx42.2^\circ$
	A1	For awrt 42.2°

Type MS · Using the sine rule, including the two-answer case

Answer 14 of 14 · Question 9 part b · 5 marks

2024 JUN · Paper 02 · extract 3 of 4 · find part b in this mark scheme extract

USEFUL SKETCH



NB:

$AC = 25.03$ cm

$AF = 17.57$ cm

$CF = 22.65$ cm

Angle $FCA = 42.9°$

Type MS · Using the sine rule, including the two-answer case

Answer 14 of 14 · Question 9 part b · 5 marks

2024 JUN · Paper 02 · extract 4 of 4 · find part b in this mark scheme extract

Question	Scheme	Marks